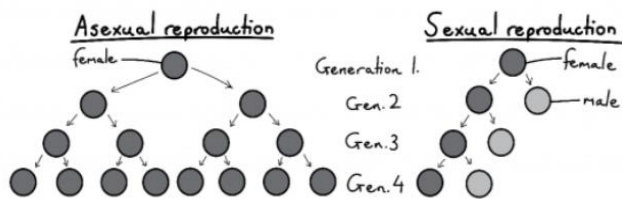
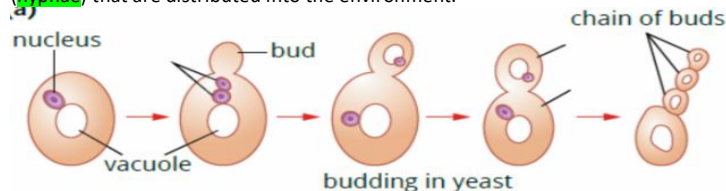


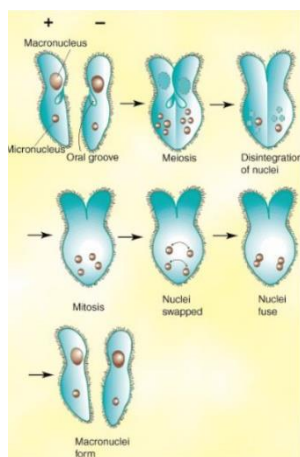
5.1.1 How does reproduction ensure the continuity of animals? Asexual- **Regeneration-(Planarian)** detached part grows into new one. **Budding-(Hydra, sponges)**-smaller buds outgrowth from parents. **Fission-(Jewel anemone, flatworms)**- produce 2 new offspring. **Fragmentation-(Planarian)**-body breaks and each one regenerates. **Parthenogenesis-(brine shrimp, water fleas)**- development of unfertilised ovum in S. **Sexual-(humans)**- gametes are produced through **meiosis**, haploid ovum and sperm, fusion of gametes (**zygote**), combination of genetic material. Genetic diversity and variation, mating. **External fertilisation-(salmon)**- the union is outside the body, large number of gametes, aquatic environment, simpler, no mating rituals. Extra energy is required. **Also frogs, corals.** **Internal fertilisation-(mammals)**-union inside the body, gametes are closer, protection of offspring, fewer offspring, mating risk of sexually transmitted infections passing btw Os.



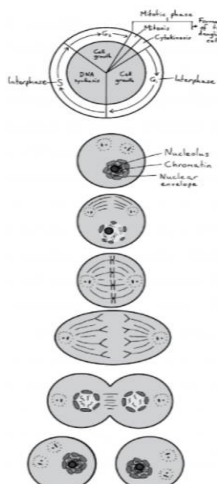
5.1.3 How does reproduction ensure the continuity of fungi? Asexual- not energy intensive, occur even under stress, may result in offspring that only suited to one habitat. **E.g. Fragmentation-(Hyphae)**- pieces of it become new colonies. **Sexual-** increase genetic variability in S. E.g. **plasmogamy**-two genetically different cells fuse together. Also **conjugation**. **Karyogamy**-the two-nucleus fuse together. **Budding-(yeast)**-asexual reproduction, the nucleus divides and bulge forms, then **cytokinesis**, then the bud detaches itself from mother. The bud is smaller than parent, but genetically identical. **Spores-mitospores-asexually-mould**, haploid cells, identical to the parent cell, large numbers, a cluster of spores can be produced inside a **sporangium**, which can be dispersed into the environment by wind or vectors. **Meiospores-sexually-ascomycetes**- gametes are generated which produce spores (**hyphae**) that are distributed into the environment.



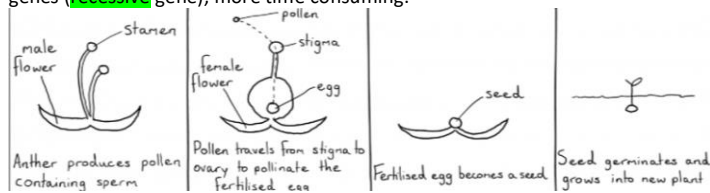
5.1.5 How does reproduction ensure the continuity of protists? Asexual- produces in large numbers, fast, only need one O, less energy, lack of genetic diversity, higher chances of pathogenesis. **Sexual-(malaria, plasmodium)**-for haploid protists, two haploid cells fuse to form **zygote**, genetic material are combined in a new, fused nucleus, meiosis occurs. For **diploid** protists, adult cells undergo **meiosis** to produce 4 gametes, gametes fuse during **fertilisation** to form diploid **zygote** to an adult. Greater variation, genes are mixed recombinantly, evolution happen. More energy intensive, slower, need more than one O. reproduces through **conjunction** under stressful environment. **Binary fission-(amoeba, euglena, paramecium)**-asexual, membrane-bound nucleus need to replicate, the body is pinched to two parts, the parental cell replaces with the two new daughter cells that are identical. **Budding-(slime moulds)**-asexual, new O grows from the body of the parent to form a new colony, it's smaller than the parent bud.



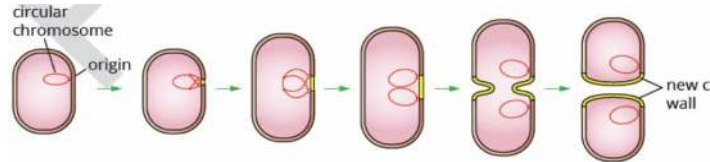
5.2.1 Mitosis-cell division, resulting in two identical daughter cells, with the same number and kind of chromosomes as the parent cell. **Interphase**-cell prepares itself for division, **DNA replication** occurs to produce two copies of each chromosome. **Prophase**-duplicated chromosome condense, the mitotic spindle forms at either end of the dividing cell. These spindles are composed of strands of **microtubules** which lengthen and shorten to pull chromatids apart, nuclear envelope breaks down. **Metaphase**-sister chromosomes line up along the equator of the cell, spindle apparatus/mitotic spindle fully formed. **Anaphase**-sister chromatids are drawn to opposite poles of the dividing cell by the mitotic spindle, **microtubules** bind to chromatids at the **kinetochore** and begin to shorten, separating pairs from each other. **Telophase**-two new nuclear envelopes begin to form around the separating sister chromatids, nuclei are visible. **Cytokinesis**-two daughter cells are separated. They are **diploid (2n)**, identical daughter cells.



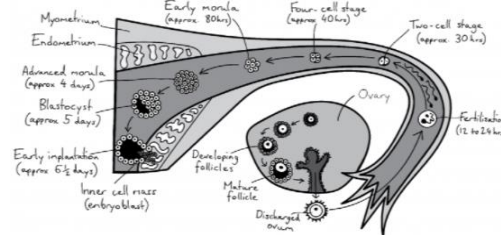
5.1.2 How does reproduction ensure the continuity of plants? Asexual-(vegetative reproduction-Bryophyte)-structural modifications to stem or root results new individuals. Favourable traits are passed, ensure consistency in production, less energy intensive, increases rapidly, exploits suitable habitats quickly. **Pathogenesis** may spread, reduction in genetic diversity, increase the susceptibility of S to new disease, reduce evolution due to lack of genetic variation. **Path-animals, parthenocarpy-plants.** **Sexual-(flowering plant-lilies Angiosperms)**-reproductive organs (flowers), **anther** produces haploid pollen, ovary contains haploid ovules, fusion of gametophytes (**pollination**) assisted by wind or insects, **fertilisation** occurs and ovules grow into seeds in a fruit, which disperse once ripe, then germinates and grow. The seed is then freed from the fruit. creates genetic diversity, higher levels of disease resistance, greater ability to adapt to changing conditions. Prevent favourable genes (**recessive gene**), more time consuming.



5.1.4 How does reproduction ensure the continuity of bacteria? Asexual-(binary fission)-produces in large numbers and very rapid in right conditions, only requires a single O to produce offspring. Lack of genetic diversity, lower chance of O survival. However, this may be overcome by high rates of **mutation** during **DNA replication**, and horizontal gene transfer (**HGT**) by **plasmid**. **Sexual-** genetic variation, adapt to the environment. **Conjunction** the genetic material will be transferred BT bacteria, as one bacterium connects to the other through a tube called **pilus**. **Transformation**- a bacterium absorbs **DNA** from its surroundings and DNA released by other bacteria. **Transduction**- **bacteriophages** work as vectors for gene transfer from one host to another. **Binary fission-(E. coli)**-asexual reproduction, single cell divides into two identical daughter cells, **DNA replication** occurs, genetic information is copied and divide in two, **cytokinesis** occurs, produces daughter cells with identical genomic information.



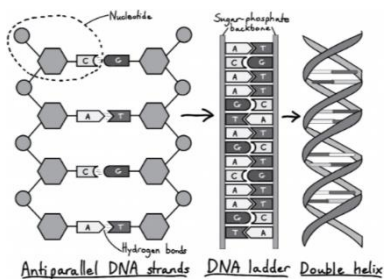
5.1.6 Features of fertilisation, implantation, hormonal control (pregnancy, birth) **Fertilisation**-the fusion of haploid **gametes** to form diploid **zygote**. It contains the genetic material. The sperm use acrosome to dissolve **zona pellucida**. The molecules on the sperm surface bind to receptors on the egg's cell. Egg surface changes. **Implantation**-when fertilised egg adheres to the wall of the uterus. First stage is **cleavage** (zygote divides to many smaller cells by mitosis). This transitions the cells to **morula**, then **blastocyst**. **Gastrulation** is where the gastrula becomes an **embryo**, then **fetus**. **Hormonal control in Pregnancy**-hypothalamus releases GnRH which produces LH and FSH. **Oestrogen**-thickens **endometrium**, produced by either a mature follicle or the **corpus luteum**, **progesterone** stabilises **endometrium**, produced by the corpus luteum, **oxytocin** causes muscular contractions and the placenta to release **prostaglandins**. **HCG** maintains **corpus luteum**. **Hormonal control in Birth-oestrogen, progesterone, FSH** stimulates **follicles** to grow in the ovaries, **LH** causes the egg to burst out of a mature follicle (**ovulation**), and form **corpus luteum**.



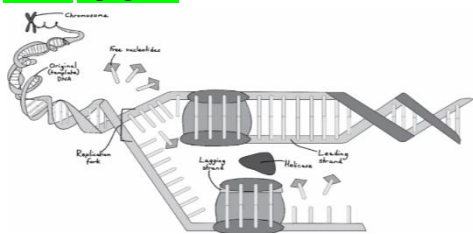
5.2.1 Meiosis-cell division, resulting in four daughter cells (**gametes**) that are **haploid (1n)**. **Interphase I-DNA replication** occurs to produce two copies of each chromosome, **homologous** chromosome approach to each other. **Prophase I**-chromosome condense, nuclear envelope dissolve, homologous chromosomes pair up, **crossing over** occurs. **Metaphase I**-homologous pairs line up along the equator of separated cell-**independent assortment** and **random segregation** occur. **Anaphase I**-homologous pairs are separated, pulled to opposite ends by the meiotic spindle, sister chromatids remain attached. **Telophase I** and **Cytokinesis I**-chromosomes arrive at opposite ends of the cell, two **diploid** daughter cells are formed. **Prophase II**-chromosomes condense, nuclear envelope dissolve. **Metaphase II**-chromosomes line up. **Anaphase II**-**microtubules** separate sister chromatids and pulled to opposite poles. **Telophase II** and **Cytokinesis II**-nuclei and nuclear envelopes form, four **haploid**, non-identical daughter cells form.



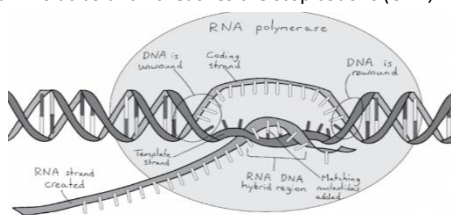
5.2.2 DNA replication, Watson & Crick model- double helix strand carries genetic information, encoded as sequences of nucleotide bases. Phosphate is bounded to sugar, forming a sugar-phosphate backbone that is bounded to **nitrogenous bases** (A, T, C, G). **Initiation-helicase** unwinds and separates complementary DNA strands by breaking the hydrogen bonds BT nitrogenous bases, moving the **replication forks** forward, the **single-strand binding proteins** prevent DNA to reform a double helix. **Elongation-DNA polymerase** directs primers to add new base pairs that are complementary to the strand in direction 5' to 3'. The **leading strand** is continuous, the **lagging strand** is discontinuous. **Termination-DNA polymerase** reaches the end, and two identical daughter strands are produced that recoil into the double helix shape. Enzymes essentially 'proofread' the double helix structures. **DNA polymerase III** makes the bulk of the new DNA, **DNA polymerase** and **DNA ligase** seal **Okazaki** fragments and repair base.



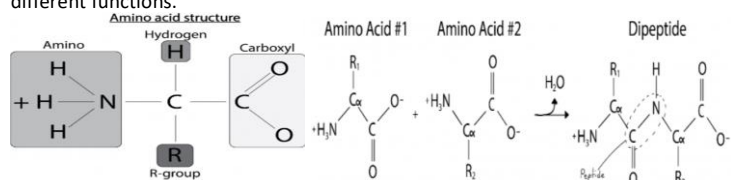
5.2.3 Effect of Cell replication on continuity of S-DNA replication- DNA is the fundamental of hereditary unit, Reproduction of cells is dependent upon **DNA replication**. Replication ensures that important information for life is transferred down through the generations. **Mitosis** and **meiosis** need **DNA replication**. DNA replication ensures that daughter **DNA** strands carry the same gene and encoding all essential proteins for life. **Mitosis**-growth and repair of Os, replace old tissues, mitosis allows us to develop to maturity when we can pass our genetic information into **offspring** through **sexual reproduction**. Some Os reproduce by **asexual reproduction**, which is facilitated by mitosis. **Meiosis**-the combination of **gametes** during **sexual reproduction** creates new Os. **Meiosis** introduces variation. **Crossing over**, **independent assortment** and **random segregation** allows for combination of different alleles, increasing variation and the wider population. Variation and **mutation**-survival and evolution. **PMAT + cell cycle**



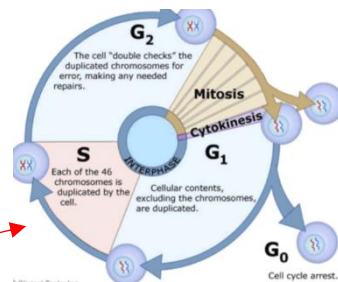
5.3.2 Transcription & Translocation- **Transcription** is the process of turning genetic information stored in the DNA into **mRNA**. **RNA polymerase** binds to the promoter sequence. A small region of DNA is unwound by **helicase**. RNA polymerase 'reads' the DNA template strand to create pre-mature **mRNA**. **Terminators** sequences end the **transcription** and pre-mature mRNA undergoes **splicing** where it removes **introns** and combine **exons**. It also adds a **poly-A-tail** to the 3' end and **cap** to the 5' to protect it and mRNA now can go through nuclear pores. **Translation**-the process of turning **mRNA** into a **polypeptide chain**. **mRNA** docks to a **ribosome**, the ribosome matches a **tRNA** molecule to the **mRNA** by matching codons and anticodons. tRNA reads the mRNA start codon (AUG) and brings the matching **amino acid**. Then, a **polypeptide chain** is formed BT the adjacent amino acids until it reaches the stop codons (UAA, UAG, UGA). The ribosome releases the mRNA and **polypeptide chain**, which then folds and become mature protein.



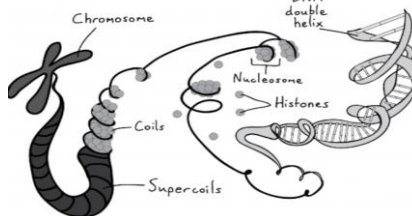
5.3.4 Function & Importance of polypeptide synthesis- **polypeptide synthesis** is used to produce proteins in cells. Proteins are super important for cell and O structure and function. Effective polypeptide is particularly important for multicellular Os, which require a high degree of coordination, facilitated by protein interactions. If protein synthesis would not work properly, **diseases** such as **cancer** may occur. Polypeptide synthesis is also important for increasing the complexity of Os. The ratio of proteins to genes increases, as Os become more complex. E.g. **E. coli** genes and proteins are 4,288 and 4,700 compared to **Homo Sapiens**, which are 20,000 and 1,000,000 This is because of each stage in the polypeptide synthesis process-**transcription** and **translation**-there are opportunities for variation. Segment of **mRNA** may be rearranged to produce different variants from a single gene, and a variety of **post-translational modifications** mean that polypeptides may be edited to perform different functions.



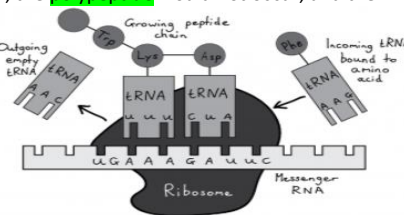
Use of models in science **DNA Replication, Protein Synthesis, Mitosis and Meiosis models**- the use of models in science is important because they explain complex ideas and make them easy to understand. E.g. **DNA replication**, **protein synthesis**, **mitosis** and **meiosis** are very complex ideas and processes because they are involved in many steps. However, thinking of them in your head and processing them by using everyday materials (pipe-cleaners, sticks, Velcro, coloured pom-poms) to model them helps the student with their understanding. Thinking about how to move parts and what to use to move these parts increase their understanding of how these processes occur. Overall, using models in science deepen your knowledge of the difficult concepts and ensures that you do not forget them, and you also see the difference BT more clearly. You can now easily compare them and know the key differences in them, you can easily demonstrate them and observe them. However, there are limitations of it because it does not show everything, and they are not up to date.



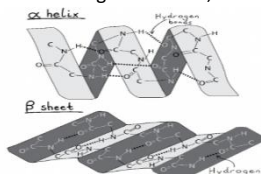
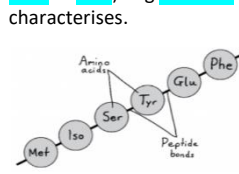
5.3.1 DNA in prokaryotes & eukaryotes- **Eukaryotic DNA**-the DNA is stored in the **nucleus**, tightly wound around **histone** proteins, coiled into supercoiled chromosomes. Large with non-coding and repetitive sequence, linear DNA, multiple chromosomes. DNA in **mt DNA, chloroplasts**. It is parallel shaped and **double helix**. They have larger **ribosomes** and have different protein of **RNA**. There are **telomeres** act as buffer. **Prokaryotic DNA**-free-floated in **cytosol**, small, compact **genomes**, circular chromosomes, unbound by proteins and extra-chromosomal segment of DNA called **plasmids**, DNA is small circular plasmids. Plasmids are to be transferred BT Os to pass genetic material horizontally within generation, single chromosome. It is parallel shaped and **double helix**. They have smaller **ribosomes** and have different protein of **RNA**. There are no **telomeres** as prokaryote DNA is circular with no ends that can be eroded with each replication process.



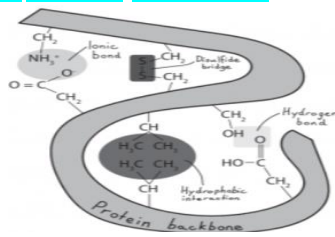
5.3.3 Importance of mRNA, tRNA- **mRNA** is an encoding sequence that is copied during **transcription** process. It carries the genetic information to express as a gene. **mRNA** is important because without mRNA molecule, the cells can not express genes and **polypeptide synthesis** would not occur as **DNA** molecule is too big to go through the nuclear pores, so mRNA is used during polypeptide synthesis to produces **polypeptide chains**. **tRNA**-it carries **amino acids** and has **anticodons** that are complementary to **codons** that are on **mRNA** molecule during **translation** process. The tRNA 'reads' the mRNA sequence and brings the matching amino acids and it continues until it reaches the stopping codons. Then, this forms a **polypeptide chain** that then undergoes **post-translational modification**, which folds and become a mature protein, so again without tRNA, the **polypeptide** would not occur, and the important proteins would not be made. Overall, both **mRNA** and **tRNA** are vital for protein synthesis.



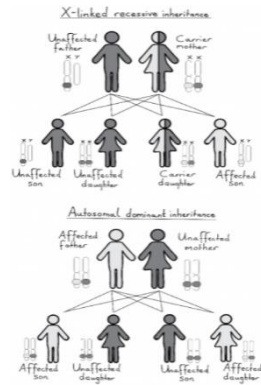
5.3.5 Structure & Function of proteins- proteins are composed of **amino acids**. **Primary structure** is when the sequence of **amino acids** that arranges during **translation**. **Secondary structure** is the formation of **alpha helices** and **beta sheets** because of **hydrophobic** interactions. **Tertiary structure** is the formation of **3D shape** as the protein **backbone** twist and bound to achieve maximum stability. **Quaternary structure** is interaction of **protein subunits**. E.g **haemoglobin** composed of **four subunits**. **How do genes & environment affect phenotype?** **Phenotype**-the outward appearance of an O. whilst an O's genetic make-up will remain static throughout its life, its observable traits may change over time because of their environment. External factors have a role in how our genes are expressed (certain genes may be switched on at different stages of development such as **homeobox** genes during **embryogenesis**), or only expressed in response to certain events (such as extreme **heat** or **cold**). E.g **identical twins**, who have exact genetic code, but develop different characterises.



5.3.6 Structure & Function of proteins in living things-structure and support: protein forms the basis of the cellular **cytoskeleton**, as well as composing important macromolecular structures such as **connective tissue**, **hair**, and **nails**. **Transport:** proteins in the cellular membrane are responsible for transferring molecules into and out of the cell. They may also carry small molecules around to body. **Storage** proteins reserve important biological materials for use in the body E.g. **ferritin stores iron**. **Enzymes:** proteins may function as **biological catalysts**, carrying out a bunch of chemical reactions inside of cell. They are used in **energy production**, **DNA replication**, **transcription** and **translation**. **Antibodies:** proteins form an important part of the immune response by recognising and binding to foreign particles. **Messengers:** Hormones are proteins which transmit signals around the body, allowing the complex array of biological processes which occur to be coordinated effectively.



5.4.2 New genotype combinations in meiosis (alleles)-Autosomal (dominant, recessive)-when traits (alleles) are passed on the **autosome**-all chromosomes except for X and Y chromosome. **Autosomal dominant**-is when a trait is determined by the expression of a **dominant** allele, never skip generation, . **Muscular dystrophy**, **Huntington**. **Autosomal recessive**-two **recessive** alleles are required to be inherited in order for a trait to be **phenotypically** expressed. Can **skip generations** and evenly affected. **Sickle cell anaemia**, **cystic fibrosis**. **Sex-linkage**-when traits (alleles) are passed on the **sex chromosomes** (X, Y). Traits may be X-linked-present only on the X chromosome or Y-linked-present only on the Y chromosome. It affects one gender at higher frequencies. **Haemophilia**. Females can be **carriers** (unaffected if they have a **dominant** allele but have the **recessive** gene encoding the trait). Males cannot be **carriers** as they do not have equivalent gene on the Y chromosome to override it. So, they will be always affected if they have the trait



5.4.3 Interpreting data from-Punnett Squares-diagram used to predict the genetic outcome of **sexual reproduction**. Punnett squares are super useful tools that allow us to theorise how alleles for genes may be passed from parents to **offspring**. They may be used to calculate the probability of certain **phenotypes** based on predicted **genotypes**. Identify the **genotypes** of the parents, draw a square with four boxes, split up each parent's allele to align one with each box, fill in the Punnett square, keeping in mind whether the alleles are **dominant** or **recessive** and if the parent is **homozygous** or **heterozygous**. **Mendelian ratio is 3:1**. Interpret-what are the probabilities of **offspring genotypes**? What are the probabilities of **offspring phenotypes**? What are the ratios of different **genotypes** and **phenotypes**? Also be aware of the type of inheritance (**Normal**, **Sex-linked**, **incomplete dominance**, **co-dominance**, **Dihybrid**). **Mendel's findings:** each parent donates one allele for every gene to their offspring. Some alleles are **dominant** and expressed whether is single or two copies.

Probabilities of offspring genotype:

- 50% Pp: heterozygous purple
- 50% pp: homozygous white

Probabilities of offspring phenotype:

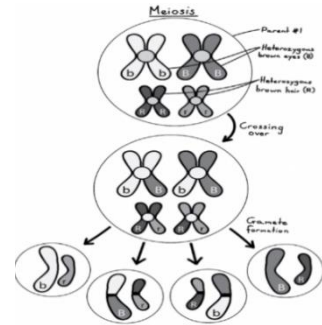
- 50% purple
- 50% white

	P	p
P	Pp	pP
p	pP	pp

5.4.4 Frequency data for characteristics in a population-the inheritance of **ABO** blood group alleles is a mix of both simple and **co-dominant**. There is one gene for the presence of antigen (A, B, AB, O) and a gene for rhesus factor (positive or negative). The A and B alleles are **co-dominant** over the **recessive** O allele and rhesus positive is **dominant** over rhesus negative. Possible **phenotypes** are (A, B, AB, O), possible **genotypes** are (AA/Ai, BB/Bi, AB, ii) and possible rhesus are + or -. Current statistics from the Red Cross indicate that around 40% of Australians are type O+, and type AB- is the rarest at less than 1%. Indigenous Australian have high frequencies of A blood groups, and it is believed that B groups were absent from population until European settlement. **Height** is **polygenic** meaning that the **phenotypes** are a result of multiple alleles. The frequency of height occurs in a curves bell, on average males are taller than females. The sample of **year 12** biology class-the most common height range was 169, the least common were 158, 185, 182.5. this only represent a small class in a small town. Representing females and males separately will improve the data.

Blood type	Frequency in Australian population
O +	40%
O -	9%
A +	31%
A -	7%
B +	8%
B -	2%
AB +	2%
AB -	1%

5.4.1 Genetic Variation in Meiosis-Crossing over (homologous chromosomes)-variation is introduced during **meiosis**, **crossing over** of **homologous chromosomes** creates new combinations of chromatids occurs at **chiasma**. **Gamete formation** sorts of chromosomes independently of one another, meaning that a number of different chromosome combinations may be formed. **Fertilisation**-during **fertilisation**, there is further opportunity for variation to be introduced. Alleles from parent one combine with those from parent two to restore a full set of chromosomes. This may result in generation of different **offspring genotypes** and **phenotypes**. **Mutations**-**Deletion**-when a part of a chromosome is left out. **Insertion**-when part of a chromatid breaks off and attaches to its **sister chromatid**, results in a duplication of genes on the same chromosome. **Inversion**-when part of a chromosome breaks off and reattaches backwards. **Translocation**-when part of one chromosome breaks off and is added to another chromosome. Variation and evolution.

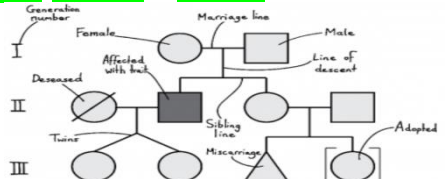


5.4.2 New genotype combinations in meiosis (alleles)- Co-dominance-when both alleles in a gene pair are fully expressed, e.g. **roan cow** and **ABO** blood group system can involve alleles for blood type A, B both being **dominant** when an offspring inherits an A allele from one parent and a B allele from another parent, both are expressed **phenotypically** in the AB blood type. **Incomplete dominant**-when the phenotypes are blended, but genotypes are not **snapdragon**, **different low pitch of human voice**. **Multiple alleles**-inheritance where three or more alleles exist for a single trait. There may be multiple alleles within the population, multiple variation of these genes may display different patterns of **dominance**. **Rabbit fur inheritance**-the gene for fur colour, C, has four different alleles, C-black, C^{ch}-chinchilla, C^h-Himalayan, c-albino. There is a pattern of **dominance** BT these different alleles: C is **dominant** to all other alleles. C^{ch} is **dominant** to c^h and c, but **recessive** to C. C^h is **dominant** to c, but **recessive** to C and C^{ch}. c is **recessive** to all other alleles. Inheritance patterns will be determined by the interactions of these different alleles.

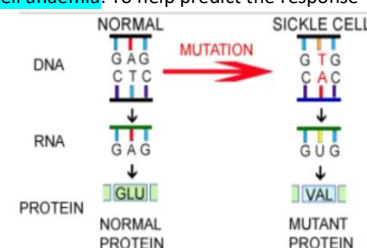
Genotype			
CC	C ^{ch} C ^{ch}	C ^h C ^h	cc
Phenotype			
BLACK	CHINCHILLA	HIMALAYAN	ALBINO

5.4.3 Interpreting data from-Pedigrees-charts displaying the **phenotypic** characteristics of Os across generations. Pedigrees are used to show how traits are passed within families, using standard symbols. Individuals in each generation should be on separate lines, level with one another, and labelled (1, 2, 3, ...). Females are always represented by circles and males by squares. A key must always be included and should explain how the chart is coloured-affected individuals are shaded black, unaffected individuals are white. Lines are drawn to indicate relations. Interpreting-tracing the passing of certain alleles from parent to offspring. For each gene there may be a number of alleles, and that two may be inherited by each individual. So, identify whether the parents are **homozygous** or **heterozygous** for a trait. Can identify whether traits are **dominant**, **recessive**, **autosomal**, or **sex-linked**. Display large data sets in a single image.

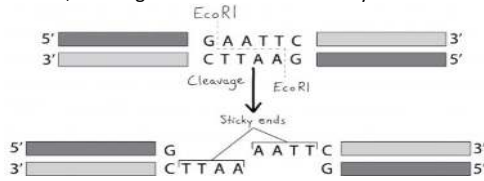
Other alleles are **recessive** and are only expressed when they are not paired with dominant gene. **Mendelian ratio 3:1**



5.4.5 Single Nucleotide Polymorphism (SNPs, SNiPs)-a change of a single nucleotide of a specific position on the **genome**. This may be a **substitution**-changing A for G, **insertion**-adding a new nucleotide, or **deletion**-removing a nucleotide. **SNPs** account for more than 90% of all differences across the human population. They occur all throughout our **genomes**, on average once every 300 nucleotides, however, they are found most within **non-protein coding DNA**. There are round 85 million SNP variants across individuals. This shows the potential for **SNPs** to drive human variation, and therefore evolution. Different **SNPs** have been identified in population, regions, and continents, as well as occurring of global frequencies. Analysis of population **genomes** has shown that some **mutations** only in certain parts of the world. Genetic screening for the presence of a disease genes-**sickle cell anaemia**. To help predict the response an individual will show to certain drugs and environmental factors-**toxins**. Tracking human migration routes. Tracking the family inheritance of disease genes. In forensics and DNA fingerprinting to solve crimes and for evidence in court cases. **Point mutation**-



5.5.1 Inheritance patterns in a population-DNA sequencing-allows us to find **single nucleotide** information for entire human **genomes**. Each individual with have a different genetic code, and these can be compared using modern computation techniques to determine patterns of inheritance through generations. By compiling large amounts of sequencing data, we can model the changes in the frequencies of genes and alleles in populations over space and time. **Sanger Method**-four **PCR** mixes are set up, each containing stocks of normal nucleotides plus one **dideoxynucleoside** (ddA, ddT, ddC or ddG). **PCR** will generate over 1 billion DNA molecules, each **PCR (Polymerase chain reaction)** mix should generate all the possible terminating fragments for that particular base. If a distinct radioactive or fluorescently labelled primer is included in each mix, the fragments can be detected by automated



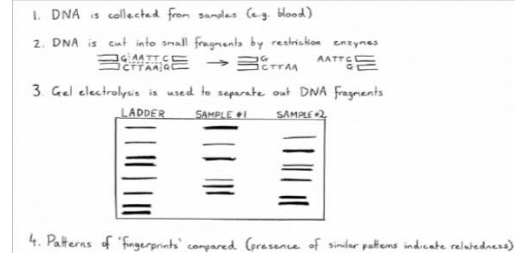
sequencing machines.

5.5.2 Using population data in conservation management-**conservation genetics** is a field which combines knowledge and approaches from population and molecular genetics with ecology and biodiversity sciences. In order to identify and propose strategies to protect S or variants at risk of extinction (preserving genetic diversity) an example of a large-scaled projects which has used population genetics to guide future conservation management was **The state of the World's Animal Genetic Resources for Food and Agriculture**, published by United Nations FAO. The report, first published in 2007 and updated in 2015, used data on population genetics across S to give an estimate of conservation status. This allows the international community to understand the current state of S stability and adapt appropriately in order to conserve diversity. E.g. **Tasmanian Devil facial tumours disease**. Scientists developed ways to "switch on" the Tasmanian devil's immune system, so it fights the cancer. **Total number of the population ÷ Number of times an allele appears, 2 white snap dragons in a population of 20. The frequency is $20 \div 2 = 10\%$. Plant – ground orchids endangered.** Cycad plants were analysed for their genetic variation using the Amplified fragment length polymorphism (**AFLP**) characterisation technique which is able to identify **genetic markers**. The AFLP technique is a type of DNA fingerprinting (or **DNA profiling** technique). It is useful because it can help identify the similarities or differences between the genetic marker on each plant. Thus, AFLP allows the identification of any **polymorphisms** in cycad plants' genome.

5.5.3 Pop'n studies - Inheritance of a disease or disorder-**Alzheimer's** the large-scale collaborative project called the International Genomics of Alzheimer's Project (**IGAP**) led to a breakthrough in discovering the more than 20 new genes as possible causes of the late development of Alzheimer's disease (AD), i.e. people after the age of 65. Each of these possible causes are related to a gene located at different locus on a chromosome. A total of more than 50,000 patients that have perceived risk of developing OR have developed AD is now part of the IGAP's database. The identification of more than 20 new genes as possible causes of late development of AD was performed using genome-wide association study (**GWAS**). GWAS that employs DNA sequencing to determine exact DNA sequence which we have discussed. This is an observational DNA sequencing study used to identify variants in DNA sequence or, more specifically, **SNPs** between people who have AD and do not have AD, affected individuals would have SNPs that occur more frequently amongst them than compared to individuals that are not affected. A typical GWAS explores thousands of SNPs. After identifying various SNPs that may be suspected to be related to late-stage AD, these varied DNA sequences are then tested for statistical significance using meta-analysis. Upon analysis, these genes that researchers suspect to be related to late development of AD can be classified into different groups. These genes associated with the development of late-stage AD were grouped based on being responsible for the immune response, **lipid decomposition** and synthesis as well as **recycling vesicles**.

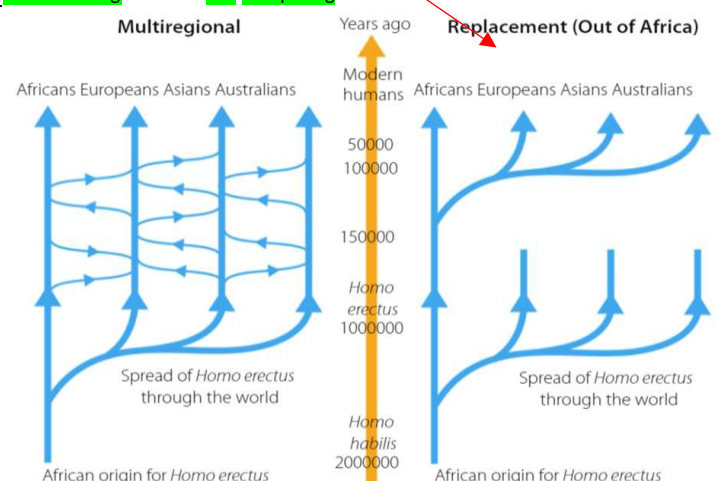
5.5.4 Population genetics relating to aboriginal migration-by analysing **mtDNA haplogroups**, we have been able to trace evolution of the human race back to our 'mitochondrial Eve', the matrilineal most recent common ancestor (MRCA), who probably lived about 200,000 years ago in Africa. An example of recent research in this area in 2017 on which research in this area in 2017 on which researchers from Australia and a number of different countries around the world collaborated, entitle; **'Mitochondrial DNA diversity of present-day Aboriginal Australian and implications for human evolution in Oceania.'** By analysing **mtDNA**, researchers have been able to identify lineages of Aboriginal Australians. A high degree of genetic diversity was found across the continent, and various ancient **haplogroups** supports the hypothesis that ancestors of Aboriginal Australians entered through at least two entry points. **mtDNA** data also support the hypothesis of long-term isolation of Australian continent.

5.5.1 Inheritance patterns in a population-DNA profiling-also known as **DNA fingerprinting**, is a technique which allows scientists to determine an individual's unique DNA characteristics. This technique is used widely in forensics. It can also be useful to identify patterns of inheritance BT individuals. The technique is effective because human DNA contains large stretches of **junk DNA**, which vary in length and contains different numbers of repeats. These are called **variable number tandem repeat sequences (VNTRs)**. The individual variability within these sequences allows us to generate individual 'fingerprints' from our DNA. Collection of DNA samples, cut out DNA, using restriction enzymes (e.g. **EcoRI** cuts **GAATTC**). This creates a mixture of DNA fragments. **Gel electrophoresis** separates DNA fragments. The gel is visualised to generate an image of the fragments separated into bands. Patterns of bands are compared.



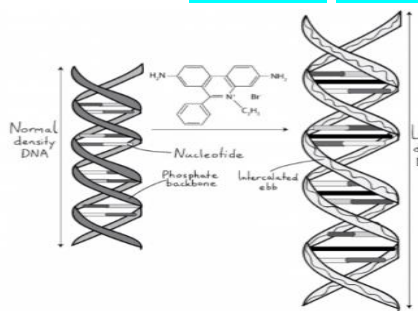
5.5.3 Pop'n studies - Inheritance of a disease or disorder-there is an approximate 0.8% nucleotide base pair variance among human individuals. Whilst this represents a relatively small difference BT people, improvements to computational technology and bioinformatics have allowed scientists to study these variations. **Haplotype**-a group of alleles inherited together from a single parent. These alleles are tightly linked in a cluster on certain chromosomes, meaning that they are very likely to be inherited together or 'conserved'. **Haplogroup**-a group of similar **haplotypes** which share a common ancestral **single nucleotide polymorphism (SNP)**. By sequencing large sets of populations, scientists have gathered information on how specific DNA sequence are passed down through generations and have determined different **haplotypes** and **haplogroups** that exist globally. E.g. **Sickle Cell Anaemia**-shows how the incidence of a genetic disease in a particular population can be related to the different factors. The international **HapMap Project** is a collaborative project undertaken by researchers around the world which aims to develop a **haplotype** map of the human genome to describe common patterns of genetic variation, such as the frequency and distribution of **single-nucleotide polymorphism (SNP)** in our global population. This may help us to understand how diseases or disorders are inherited across populations. This disease is common among people with ancestors from Africa and Mediterranean countries. The **heterozygotes** are usually healthy, and they have a trait that provides some immunity against **malaria** because it lowers the density of the **parasite**. The **protozoan** that causes **malaria** spends part of its life cycle in RBCs.

5.5.4 Population genetics relating to human evolution-different cultural groups are often linked by the prevalence of certain **haplotypes**. By mapping **haplotypes** globally, we can trace the movement and evolution of the human S from its ancestors. E.g. **tracing human mitochondrial DNA haplotypes**. Each **haplogroup** is defined by differences in human in human **mitochondrial (mtDNA)**, which is inherited only from the mother ovum. This means that sequencing **mtDNA** allows us to trace our maternal lines. **Anthropological** genetics combines DNA analysis with **historical, archaeological, and linguistic** evidence to determine the pathways of human evolution. The evolutionary forces of **mutation, natural selection, genetic drift, and gene flow** are responsible for the patterns of diversity in human populations. They have not discovered whether the human come from Africa 200 000 years ago, or did they come from a subgroup once they were out of Africa? The **multiregional hypothesis** – relies mostly on **fossil** evidence, suggests that all human populations can be traced back to when **Homo erectus** first left Africa, about 2 million years ago. It is suggested that there was gene flow btw neighbouring populations and that once they disperse into other portions of the old world, they slowly evolved into modern humans. The **replacement hypothesis** – (aka **Out of Africa or Eve hypothesis**) suggests that **archaic Homo sapiens** left Africa. It proposes that a 2nd migration out of Africa happened about 100 000 years ago and that modern humans of Africa origin conquered archaic groups and replaced them by **interbreeding** with and **out-competing** them.

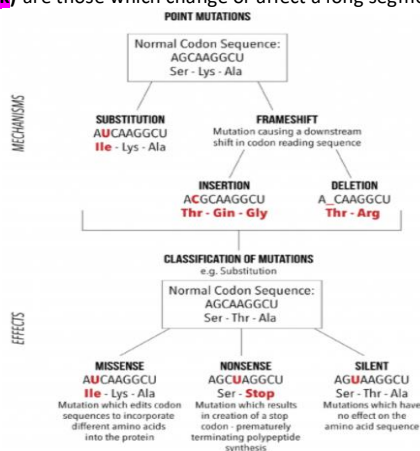


6.1.1 Mutations – EMR sources-(gamma rays, x-rays and ultraviolet (UV) light. **Electromagnetic radiation** or electromagnetic waves are emitted by electrically charged particles. These waves can interact with and ionise atoms which they encounter in the body. **Radiation** is highly penetrative because it is composed of high-energy waves. It can enter cells from external sources and interfere with DNA molecules in the nucleus. This interference can cause bonds within the DNA structure to break. When this occurs, there may be a change to the chemical composition of the DNA molecule. Parts may be deleted or rearranged, or the shape of the molecule may change. This may lead to a mutation if the DNA repair system is unable to repair the change or repairs it incorrectly. The high energy states of these radiation waves mean that they can disrupt the **hydrogen bonds** btw **nitrogenous bases** as well as bonding of the **sugar-phosphate back bone**. Radiation may damage DNA indirectly by ionising molecules in the cell to produce free radicals. These are highly reactive because they have a set of unpaired electrons which want to strip electrons from any molecules they encounter. These may react with DNA to change it. **UV light** in form of the radiations can be harmful in high doses. Excitation of adjacent **pyrimidine** (T or C in the genetic code) by UV induces the formation of **covalent linkages** (**pyrimidine dimmers**). **Pyrimidine dimmers mutations** are the main cause of **melanoma**. **Page 8**

6.1.3 Mutations – naturally occurring mutagens-there are also a number of natural environmental factors which can create mutations in DNA. **Viruses-HPV** causes **cervical cancer**- replicate by inserting their DNA into **host** cells. This creates a disruption in normal cell function and may lead to lasting mutational changes. **Bacterial infections** can induce **inflammation**, which may reduce the efficiency of DNA repair systems, increasing the rate of mutation. This is the reason that **infection** from **Helicobacter pylori** has been linked to development of **stomach cancer**. **Transposons** or 'jumping genes' are segments of DNA which can change their position in chromosomal DNA. This can create mutations which alter **gene expression** in cells. They are responsible for **phenomes** such as **multi-coloured maize**, **colouring in peppered moths** and **conferral of antibiotic resistance in bacteria**.



6.1.5 Chromosome mutation(block)-are those which change or affect a long segment of DNA (i.e a significant portion of a chromosome). Types:
Deletion: when a section of chromosome is removed.
Inversion: when a section of a chromosome is inverted (turned upside down) and re-inserted into the chromosome.
Translocation: when a section of a chromosome is moved to a different non-homologous chromosome (e.g., a portion of chromosome 21 is moved to chromosome 22). **Duplication**: when a section of a chromosome is doubled. **Down syndrome**. **Page 8**



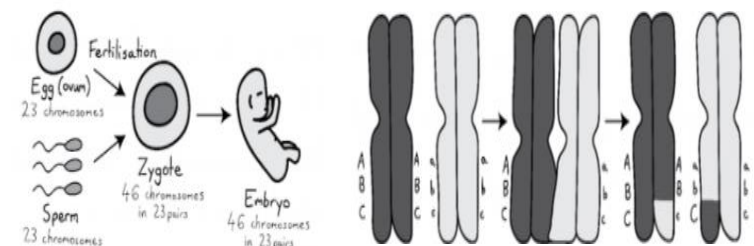
6.1.7 Coding & Non-coding DNA segments in mutation-**Codon DNA**: sequences of DNA which encode for proteins (genes are coding DNA). **Non-coding DNA (junk DNA)**: sequence of DNA which do not encode for proteins. There are an estimated 19,000-20,000 protein-coding genes, only about 1.5% of entire genome. The rest of the human genome is classified as "non-coding" DNA. **Exons**: a chunk of DNA that makes it into the final mRNA, it is a coding segments of the nucleic acid that are involved in gene expression. E.g Mutations in **tumour suppressor** genes can result in **cancer**, as can mutations in genes that activate proto-oncogenes. People who have **Xeroderma pigmentosum** are extremely susceptible to cell damage if exposed to UV light and are a thousand times more susceptible to **skin cancer** than people who don't have the **XP** mutation. Exon shuffling is a process that involves exons duplicating or moving to create a new intron-exon sequence. This can form a new gene and lead to a new protein with a new combination of functions. Exons often duplicate and reposition in the genome. **Introns** (discontinuous or split genes)-only in eukaryotic Os. Usually have no effect because don't translate. Mutation can cause incorrect splice out of mRNA. This will have effect on gene expression because the introns will be translated into the protein. Mutations in introns can lead to developmental abnormalities in embryos, eg: cause an isolated (**non-syndrome**) **congenital heart defect**. **Predisposition** of specific diseases in adults, eg: **heart disease**, **diabetes**, **cancer**, **obesity**. Acting as a **buffer** area dividing 1 gene from the next, hopefully minimising the mutation to the intron and not the exon.

6.1.2 Mutations – chemicals-(ethidium bromide reacting with DNA) there are many chemicals which interact with and have a **mutagenic** effect on DNA, both directly and indirectly. **Radioactive agents** such as **uranium**, release radiation in term of **alpha** and **beta** particles (as well as **gamma** waves). These can penetrate the cell and interact with DNA bonding to create disruptions. Interacting agents, such as **ethidium bromide**, have been used widely in the field of molecular biology to visualise DNA during experiments. These chemicals insert or 'intercalate' themselves btw the **nitrogenous bases** of DNA, resulting in a colour change to the molecule while this makes them great for identifying DNA in the lab, it can cause **frameshift mutations** during DNA replication, making them highly **carcinogenic** in the human body. **Metals**-(**arsenic**, **nickel** and **cobalt**) have **mutagenic** effects through a number of different mechanisms. They have been shown to effect processes in DNA repair, such as the ability for proteins to recognise base-pair **mismatching**. this affects the cell's ability to correct errors, thus allowing mutations to proliferate. It has also been demonstrated that some metals reduce **fidelity** during DNA replication. This means that the cell does not correctly copy the genetic code when it is replicating, so errors are incorporated into new DNA strands. **Nickel** has also been shown to inhibit the ability of **histones** to condense DNA, affecting chromosome formation. **Teratogens** - **thalidomide** (birth defects)

6.1.4 Point Mutations-are those which only change or affect one (or a few) nucleotides within a gene sequence. **Substitution**: when one nucleotide is swapped out for a different one (e.g. an A within a sequence is swapped for a C). **Insertion**: when nucleotides are added into a sequence. **Deletion**: when nucleotides are deleted from a sequence. Insertion and deletion mutations are classified as **frameshift** mutations because their effect is to 'shift' the sequence up or down so that codons are aligned. Effect: **Silent**: the mutation has no effect on the codon (due to in-built redundancy of the genetic code) and therefore, has no effect on the amino acid the sequence encodes for. This result in no change to the polypeptide or protein. **Missense**: the mutation affects one codon, introducing a different amino acid into the polypeptide sequence. **Nonsense**: the mutation prematurely introduces a stop codon, resulting in a shortened polypeptide chain, and thus a dysfunctional protein. **Point mutations** are the cause of the most common types of genetic variation in human. **Genome**: **single nucleotide polymorphisms** (SNPs), SNPs are differences single nucleotides in genes, and they account for more than 90% of all differences in the population. Disease such as **Sickle-cell anaemia**, **beta-thalassemia** and **cystic fibrosis** are all the result of SNPs.

6.1.6 Somatic mutations-are genetic alterations which a cell acquires, which may then be passed on to daughter cells by cell division (**mitosis** in humans). Somatic mutations are usually caused by environmental factors such as the **external mutagens**. These mutations affect any cells descended from the original cell in which the mutation occurred. As the affected cell divides, a specific area of tissue with mutation may develop, but the mutation will not alter the genetic composition of other cells in the body. Somatic mutations are not passed onto offspring. Many diseases such as **cancer**, are the result of somatic mutation.
Germ-line mutations-are mutations in the germ cells (sperm or ovum), which may be passed onto offspring during fertilisation. Germ cells or gametes are the basis of all other cells in the body. When fertilisation occurs, a paternal gamete and a maternal gamete combine to form a zygote. This zygote undergoes mitosis to produce all cells in the body. Therefore, any alteration to the genetic composition of the gamete will be passed onto all cells of offspring which inherit this information. Germline mutations are caused by a variety of factors, both internal and external. Diseases such as **Sickle-cell anaemia**, **cystic fibrosis**, and **colour blindness** are the result of germ-line mutations. **Page 8**

6.1.8 Causes of genetic variation-Fertilisation. Two gametes (one sperm and ovum) come together to form a zygote. The full set of chromosomes is restored (23 pairs, 46 in total). The variation occurs through the random selection gametes and interaction of dominant and recessive genes. **Meiosis**-the production of gametes (sperm and ovum). One parent cell becomes four daughter cells, each with half the number of required chromosomes (one chromatid of each chromosome). Variation occurs through mutation during **DNA replication**, **crossing-over**-may go wrong, **random segregation** and **independent assortment**. non-disjunction homologous may not separate. **Mutation**-there are several mutations that lead to variation-EMR source, chemical, naturally occurring mutagens, point mutations and chromosomal mutations. **Above**



6.1.8 Gene pool of populations: **Mutation**-the process that create new alleles in a population. Initially mutation increases the variety of **alleles** in the **gene pool**. It is the most significant cause of change to gene pool mutation may be positive, negative, or neutral. **Gene flow**-transfer of genetic variation (different alleles) from one population to another. Gene flow describes how the migration of individuals from one population to another will result in **transference of alleles** into and out of populations. This will result in a change to the frequency of alleles, changing the distribution of genetic diversity. E.g. **interbreeding programs in different Tas Devil populations**. **Genetic drift**-when the relative frequency of alleles within a population change, due to the disappearance of particular genes as result of **natural selection**. Genetic drift may be as a result of a variety of factors and may occur slowly or quickly depending on the types of selection pressures applied to a population. Two causes of genetic drift include **bottlenecking** and the **founder effect**. E.g **evolution of cheetah, Tasmania Devils**. **Bottlenecking**: when some sort of catastrophe or major event reduces a large population dramatically to produce a much smaller population. **The founder effect**: the impact of initial immigrants or **original founders** (sources of traits) in small, isolated populations. E.g. **islands** e.g. **Galapagos iguanas** (first marine iguanas colonised the coast, then) diversified into **2** colours. Now there is a **polar inland iguana**-so **1** founder to **3** sub-S.

6.2.1 Biotechnology – changes to biodiversity-Present – animal- **'pharm' goat**-goats naturally produce high volumes of milk and have relatively short generational times (18 months). These attributes have led to them being chosen as **'bioreactors'** to produce helpful proteins for humans in their milk. **ATryn** goats are **transgenic** animals used to produce **pharmaceuticals** (medicines) for human use. ATryn goats have had the human gene for **antithrombin-III** (AT-III) inserted into their genome. So, in ATryn goats, the human gene for AT-III is inserted alongside a goat milk specific promoter to ensure that the goats produce the AT-III protein in their milk. The AT-III is then isolated from the milk of ATryn goats for medical use in humans. In humans, AT-III helps to ensure that blood does not clot-e.g. genetic disease that makes them **AT-III deficient**. **Present – plant-BT cotton**(bacillus thuringiensis- Traditional **pesticides** used on cotton plants had to be made stronger caterpillar of the **Helicoverpa zea** moth-destroys cotton plants. They contain a gene that codes for a protein that kills the Helicoverpa zea moth. This protein is harmless to the environment, but when eaten by the moth it is converted in it's digestive system to the active form that kills it. Hundreds of millions of dollars' worth of cotton saved. Now farmers only spray occasionally to get rid of sucking insects using a narrow spectrum **insecticide** that does not wipe out beneficial insects such as lady bugs and wasps (**pollinators**). Increased spraying of the caterpillars increases their immunity to the pesticide.

6.2.2 Future direction of uses of biotechnology- Future directions involve **genetically modified crops** and **animals** with new breeds and varieties able to survive particular climate conditions. e.g. **drought** or **high/low temperatures** or **disease resistance**. New **nanotechnologies** are being developed. e.g. in **Nano-medicine**, **biomaterial energy production**, **bio-mimicry** and **tissue engineering**. There is also research into many medical applications such as **genetic diseases** looking for new **'Medical tourism'** and the global organ trade.

Agricultural benefits arise with increased crop yield due to **herbicide tolerance**, insect and **pathogen resistance**, nutritional enhancements and growth of crops in substandard conditions.

Bioengineered farm animals increase productivity and also help in the supply of food to the ever-increasing human population.



6.2.4 Changes to biodiversity due to genetic techniques- **Biodiversity** comprises all the millions of different S that live on our planet, as well as the **genetic** differences within S. It also refers to the multitude of different ecosystems in which S form unique communities, interacting with one another and the air, water and soil. Biodiversity is explored at three levels: **Genetic biodiversity** - variety of genes within a S. S biodiversity - the variety of S within a region. **Ecosystem biodiversity** - the variety of ecosystems in a given place. **Conservation of biodiversity**. Os and their roles are essential for the ecosystem's survival. S interactions are complex, and the loss of key S can have a substantial impact on the ecosystem. If one S disappears it could trigger a domino effect in the food chain. Use of **GM crops**: This is one of the most feared threats to the loss of **biodiversity** as more farmers use **GM crops** this will reduce the **gene pool**. If disease and environmental changes occur there can be widespread disasters. Genetic variation is important for **evolution**. Conservation of biodiversity. Maintaining habitat and 'green corridors. Maintaining S and breeding programs in zoos, botanical gardens, arboretums. Plant propagation using biotechnology to increase yield of crops. Biotechnology used to make seed banks 1,000 seed banks located all over the world About 800 miles from the North Pole on the icy and sparsely populated island of Svalbard, sits a concrete and mirrored tunnel jutting out from a glacier-encrusted sandstone mountain that overlooks an inky fjord. The **Australian National Botanic Gardens National Seed Bank** is home to a large and ever-increasing collection of **Australian native seeds** (10,000 S)

6.2.1 Biotechnology – changes to biodiversity-Past – animal-insulin- In 1978, a synthetic version of the human insulin gene was constructed and inserted into the **bacterium Escherichia coli**, in a laboratory in USA. Insulin is a protein consists of a chain of building blocks called amino acids. When the sequence of amino acids is known, the corresponding sequence of DNA can be isolated (or in this case, chemically synthesised) and introduced into bacterial cells to make the human protein. To accomplish this, the piece of foreign DNA is first inserted into a **plasmid**, a small circle of DNA which serves as a carrier. The new **"recombinant"** plasmid carrying the human gene is then reintroduced into another bacterial cell. Once inside the cell, the human gene on the plasmid can be read by the cell's protein-making machinery. **Page 8, Past – plant-Roundup Ready soybeans**-roundup is a popular brand of **herbicide** that can be used to kill weeds. However, it is a problem because it is not very selective and can also kill agricultural crops. The active chemical In Roundup is **glyphosate**, which inhibits an enzyme in plants that is normally used to synthesise amino acids. Without this enzyme, the plant cannot synthesise amino acids and hence cannot make proteins, leading to the death of the plant. Monsanto, the company who make Roundup herbicide has **genetically engineered** crops that were **resistant** to **Roundup** and transferred a gene from a bacterial S that is resistant to glyphosate into crop plant S.

6.2.1 Biotechnology – changes to biodiversity-Future – animal-GM Pig- **Xenotransplantation**: adding and removing genes with **gene editing** technology creates genetically-altered pig cells. These are used to make pig embryos. The genetically engineered pigs are raised in a controlled, bio-sealed environment. The organ is removed from adult pig and transplanted into patient. Patient must still take **immunosuppressant** drugs, to prevent their body rejecting the new organ. However, the first person in the world to get a heart transplant from a **genetically modified** pig has died. **Future – plant-GM crops-** they left restriction in Australia recently. E.g **GM safflower** in Australia-High **oleic acid** has been achieved by changing the **expression** of **two genes** already present in **safflower**. This change results in more oleic acid and less linoleic acid in the GM safflower seeds. GM safflower can only be grown under a licence issued by the Gene Technology Regulator (the Regulator). Criminal charges can apply for non-authorised dealings with GM crops of any kind. These can apply for future crops and plants in Australia.

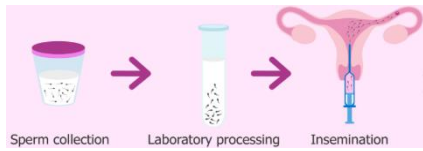
Biotechnology uses biological processes and Os to create new products and develop technologies.

Darwin's findings: natural selection leads to variation, which leads to evolution. If a gene confers an advantage, it is more likely to be passed on to the next generation.

6.2.3 Potential benefits for society of research using-genetic technologies- **Biotechnology** can have profound effects on the lives of humans and their social structures. E.g.: **Medical advances** resulting from biotech applications improve disease diagnoses and treatments, make designer drugs available and allow for individualised treatments based on a patient's specific **genetic profile**. Led to the **development** of **vaccines** and **treatments** for chronic diseases, as well as early detection of some diseases, slowing the progress of some conditions and improving prognoses and quality of life. Increasing the **nutritional** value of **foods**, which can improve health, together with greater food availability due to better crop yields. Debate continues around issues of equitable access, cost patents, privacy, consent and safety for individuals and communities. Concerns have also been raised about the ethics of using **stem cells**, **embryos** and **animals** in biotech research and development. Ethics (and ethics committees) are important where humans are directly affected (e.g. **medicine**), and where **genetically modified Os (GMOs)** such as **food crops** are concerned.

6.3.1 Genetic technologies currently inducing genetic change-reproduction technologies: **Artificial Selection** or **Selective Breeding**- human selecting Os with desirable traits in a captive population for interbreeding to produce the next generation, at exclusion of natural, increase production, E.g. **Ewes produce wool**, **Angus cows**. **IVF**-(in vitro fertilisation)- eggs are removed from the ovaries of a woman and fertilised in a laboratory with sperm provided by her partner or a donor. One or two embryos — fertilised eggs — are implanted into the woman's uterus. Fertility treatment (**humans**). Able to freeze embryo and genetic screening. **Cloning techniques:** **Therapeutic cloning**-nucleus of an egg cell is replaced with the nucleus of the body cell, egg cell is stimulated with electricity, egg grows, embryo stem cells are collected and used to treat the donor. E.g. **medicine**-stem cells can differentiate to any cell. **Recombinant DNA techniques:** **Transgenic S**-is a S that has genes injected from another S into its genetic code. E.g. **agriculture-BT cotton** and animals-**Atlantic Salmon**. **Hybridisation**-where two genetically different strains are crossed to produce offspring with more desirable traits than either parent. E.g. **liger**, **triticale** (wheat and rye). **CRISPR**-cutting specific regions on the gene using restricted enzymes on bacteria to treat genetic diseases such as **cystic fibrosis**. **Gene sequencing**-determining the order of the four bases-that make up the DNA molecule, informs genetic information that is carried in a particular DNA segment. **Medicine** and **genetic research**-identifying genetic disorders such as **Down syndrome** and understanding of evolutionary relationship, and forensic biology. It can cause injury or trauma.

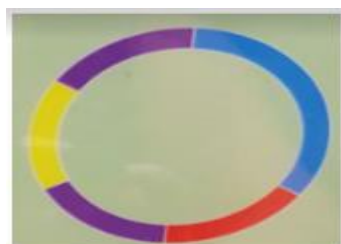
6.3.2 Artificial Insemination-(cows, horses)-involves the deliberate introduction of **sperm** into the **uterus** of an O without sexual intercourse. It has been used commonly in animal breeding, as well as in fertility treatments for humans. Sperm is collected from male genitalia, processed (it may be washed or treated with antibiotics), and then injected into the uterus of an O, where **fertilisation** occurs. The advantage of using this process for breeding is that humans can select desirable traits and forcibly combine them to hopefully produce offspring with desired characteristics. It may be used to **synchronise births** in the livestock industry or avoid injuries during mating. It also allows for **favourable genetic** material to be sent around the world (this is commonplace for race-horse breeding). 75% of dairy cattle **inseminations**, and 85% of pig **inseminations** within the agriculture industry are achieved through this artificial insemination method. A disadvantage of this process is that by selecting for the few traits we perceive as desirable; we may limit **genetic variation** within a S. An increase in **homogenous** populations may cause issues for S survival in the long term. Cost money, time consuming. Reduction in diversity.



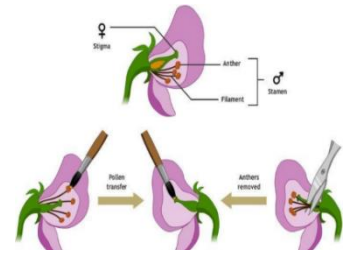
6.3.4 Whole O cloning-(banned in Australia)- creation of a new molecular O that is genetically identical to its parent O. **somatic cell** nuclear transfer. E.g. **Dolly sheep**, **embryo trials**. Livestock industry, inheritance of desirable traits. An adult cell (**somatic cell**) is removed from the O you want to clone (in **Dolly's** case, it was removed from the mammary tissue in the udder). This contains their genetic material (i.e. what you want in your offspring). An unfertilised egg is removed from a donor O. The DNA is removed from this egg – it is **denuded**. Basically, it becomes an empty vessel. The **denuded** egg and the stem cell are fused. The resulting cell is cultured so that it begins to divide and become an embryo. The **embryo** is implanted into a **surrogate** O. The surrogate gives birth to an O which is genetically identical to the donor. In terms of the effectiveness, the 'clone' produced is not strictly identical to its parent O. Since we are using somatic cells as the source of genetic material, any **mutations** acquired in that cell will be passed down (**CC** and **Rainbow** have the identical gene, Rainbow is calico in colour while CC is a tiger tabby and has no orange). Furthermore, mitochondria present in the cytoplasm of the donor egg contains DNA which is passed on to the cloned O. Therefore, the clone has a different mitochondrial genome from its target parent O. Environmental factors also have an impact on how our genes are expressed (e.g. **different phenotypes** in identical twins). Hence, a genetic clone will not necessarily grow to be phenotypically identical to its parent O. **Cloning** is also a very expensive and time-consuming process, so its effectiveness is limited. **Plant propagation- roses**. Expensive, reduces diversity and genetic variation.

PCR- Original DNA mixed with **primers** (small segments of original DNA), heat to 95° C to **denature** (open DNA), enzyme added (**Taq polymerase**) which makes the primers anneal (Stick) to original DNA, nucleotides added to mixture and heated to 72° C, **Taq polymerase** extends new DNA, two new copies formed-1st cycle, repeated-process is exponential. **Polymerase Chain Reaction** to amplify or modify DNA sequences – harder. This is by far the most successful method of amplifying (making many copies of) DNA sequences. The process is done in a test tube: the DNA required is isolated, fragmented with **restriction enzymes** and separated by **gel electrophoresis**. The DNA fragments are denatured by heating to about 95 degrees C. These single stranded DNA are exposed to a solution containing a **radioactive DNA "probe"**. The probe consists of single stranded DNA or RNA, with a sequence chosen to base pair with the required DNA. With correct temperature of about 50 – 65 degrees C, salt and correct pH the probe will bind to its corresponding sequence of the target DNA and nowhere else. As the process proceeds the DNA doubles after each cycle. **PCR** had been used to indicate the **presence** of **HIV infection** and has been used to amplify degraded DNA for use in **forensic science**. PCR occurs in a thermal cycler and involves a repeat procedure of 3 steps: **Denaturation**: DNA sample is heated to separate it into two strands. **Annealing**: DNA primers attach to opposite ends of the target sequence. **Elongation**: A heat-tolerant **DNA polymerase (Taq)** copies the strands **COVID**

Plasmids-insulin-Online databases that have the whole genome sequence, take human's DNA from one of their cells and find the insulin gene and make copies, we can work backwards from the protein structure and create genes that will make our insulin. Build **plasmid** using pipe cleaners-fast and simple chromosome: blue represents the **normal plasmid**, red represents **antibiotic resistance**-important for the survival of bacteria, purple represents the **marker gene**, which informs us that the plasmid has our gene, If the gene is not split up (the insulin gene is not in the plasmid), the bacteria will be blue in colour. If the gene is split in two (the insulin gene is in the plasmid) the bacteria will be white in colour. Yellow represents the **insulin gene**.

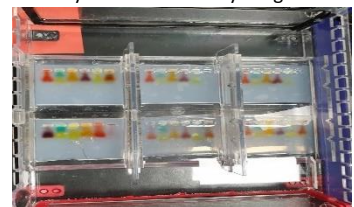


6.3.3 Artificial Pollination-when **pollen** is purposely taken from one plant and placed on the **stigma** of another plant. Artificial pollination is used in agriculture to influence the traits passed to subsequent generations of plants. It is also known as **mechanical pollination**, as it involves manually transferring genetic material from one plant to another. This technique was famously used by **Gregor Mendel** in his **pea-plant** experiments to trace the inheritance patterns of visible traits. In modern agriculture, it is used to ensure that all plants within a crop are pollinated and can produce fruit. This increases crop yields and profitability. It is also used to create new S of plants with desired characteristics. The disadvantage of artificial pollination is that it may create **monocultures** in which there is very little genetic variation. This may create S which are vulnerable to sudden environmental changes. E.g, the **Irish Potato Famine** (1845–1849) was the result of a single strain of potato blight. Creation of **monocultures** also mean that natural varieties of plants are lost due to competition, and biodiversity is reduced. Reduction in **bee** population. **Hybrid plants-variability**

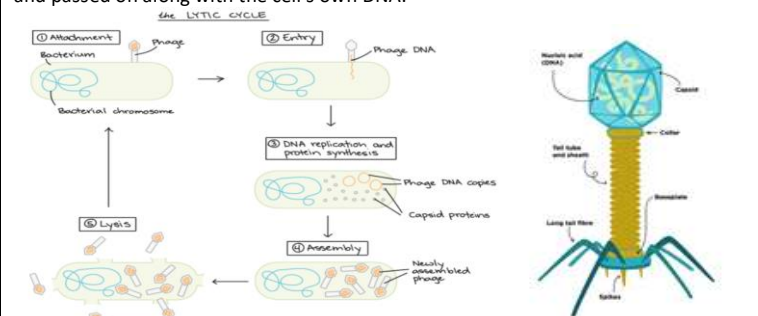


6.3.5 Gene cloning-producing **multiple identical** copies of a specific DNA sequencing. Intake a donor cell and egg cell and then fuse them together to produce an embryo. E.g. **tissue culture-orchids**. Production of biological relevant proteins (e.g. **insulin**, **enzymes** for industry). The aims of gene **cloning** are to assemble recombinant DNA containing a gene of interest, and then direct this DNA into a host O which will replicate the gene and produce the target protein in large amounts. Gene cloning can be used to synthetically produce many different proteins from many different Os, as well as to combine different proteins recombinantly. A useful target gene is identified. As now we have a huge database containing the information about gene sequences. We can send this data to companies who synthesise our desired DNA sequences for us. **Plasmids** (circular pieces of DNA) are isolated from bacteria. Both the bacterial plasmid and the DNA containing our target gene are treated using a **restriction enzyme**, which cut DNA, breaking hydrogen bonds in the molecule. These enzymes create 'sticky ends' sequences of overhanging single stranded DNA. The complementary sticky ends of the target gene and the plasmid come together through base-pairing affinity. These are then annealed using **DNA ligase**. This new recombinant plasmid is re-inserted into host bacteria by a process called **transformation**. The host bacteria express lots and lots of copies of the target gene. Producing large amount of the target protein (insulin). This protein can then be extracted from the cells, purified, and used by humans. This is relatively fast and cheap when performed correctly. Hard to produce large amounts of protein on Industrial scale.

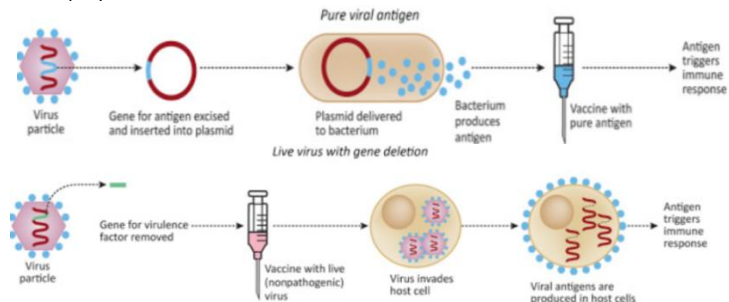
Gel Electrophoresis- We will be modelling **gel electrophoresis (Sanger Method)** using skittles colours to model proteins or chromosome fragments. Molecules of smaller sizes travel faster (e.g. yellow) than molecules of greater sizes (e.g. green). Select 5 colours, label 5 plastic cups – 1 for each colour, add 15 drops of buffer to each cup, add 3 skittles to each cup, according to colour, swirl each cup until all the colour has been extracted, add the extracted dyes to one end of your gel (wells), record the order, place it carefully in the electrophoresis unit, switch on so the current can carry the colours towards the positive end, compare the patterns produced. It snips small amounts of DNA and tests them. Some of the colours are pure except the green (blue and yellow) and mixture (brown and yellow). By using electrophoresis, you can analyse and separate **macromolecules** such as **proteins**, **amino acids**, **DNA**, **RNA** according to their sizes. This can then be used to do some tests such as **PCR** test for **Covid-19** or **paternity** tests or in **forensic**.



Phages-(small virus that only infects bacteria)- bacteriophages may use to infect their bacterial hosts: The **lytic cycle**: The **phage** infects a bacterium, **hijacks** the bacterium to make lots of phages, and then kills the cell by making it explode (**lyse**). The **lysogenic cycle**: The phage infects a bacterium and inserts its DNA into the bacterial chromosome, allowing the phage DNA (now called a **prophage**) to be copied and passed on along with the cell's own DNA.

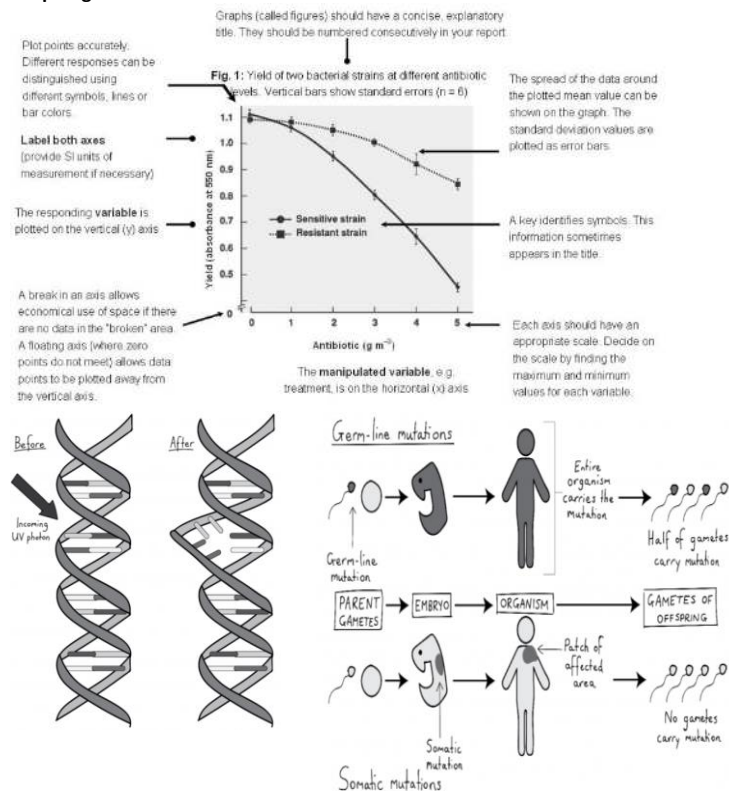


6.3.6 Recombinant DNA & Dev't of transgenic Os-Agriculture- BT Cotton – GMO: the contain a gene that codes for a protein that kills the *Helicoverpa zea* moth. This protein is harmless to the environment but when eaten by the moth it's converted in its digestive system to the active form that kills it. Hundreds of millions of dollars' worth of cotton saved. Now farmers only spray occasionally to get rid of suckling insects using a narrow spectrum **insecticide** that does not wipe out beneficially insects such as lady bugs and wasps (**pollinators**). A normal cotton seedling is cut into small pieces and grown into embryo. **BT gene** from bacteria is extracted using **restriction enzymes**. *Agrobacterium tumefaciens* acts as vector to insert genes. Cotton plant embryos are dipped in a solution of **recombinant Agrobacterium**. Embryos grown in **tissue culture**. **Medical- Viral vectors** are tools commonly used to deliver genetic material into cells. This process can be performed inside a living O (in **vivo**), or in cell culture (in **vitro**). Viruses have evolved specialised molecular mechanisms to efficiently transport their genomes inside the cells they infect. DNA can be carried by viral vectors into cells or directly inject into bloodstream or use nasal sprays.



6.3.8 Ag biotech – effect on biodiversity–The use of biotechnology can have a significant impact on the genetic variation of individuals, populations and S. A reduction in genetic variation lessens the S' resilience to changes in the environment and disease, as well as reducing the ability of the environment to meet the needs of humans in the future. **Reproductive technologies** such as **artificial insemination** (AI) and **pollination** can lead to a reduction in genetic variation by focusing on, and selecting for, fewer varieties and breeds. Eg: AI has allowed for breeding of **Friesian dairy cows** in Africa, resulting in greater milk output and economic gain. The **native Ankole cattle** are now becoming threatened. This is a concern as the breed is well-adapted to the semi-arid conditions of countries such as Uganda, and does not require great amounts of water, feed, or veterinary attention. During droughts farmers walked their Ankole cows' long distances to water sources, whereas those farmers with **Friesian breeds** lost their whole herd. On the other hand, though, AI and **artificial pollination** allow for the introduction of new genes to a population, which can improve the genetic variation of the **gene pool**. **Whole O cloning** poses a risk to biodiversity, with the potential for **multiple identical genetic copies** becoming widespread in a population resulting in a loss of natural vigour. **Transgenic** and **genetically modified S** pose a significant threat to genetic variation, with threats to agriculture and the natural environment.

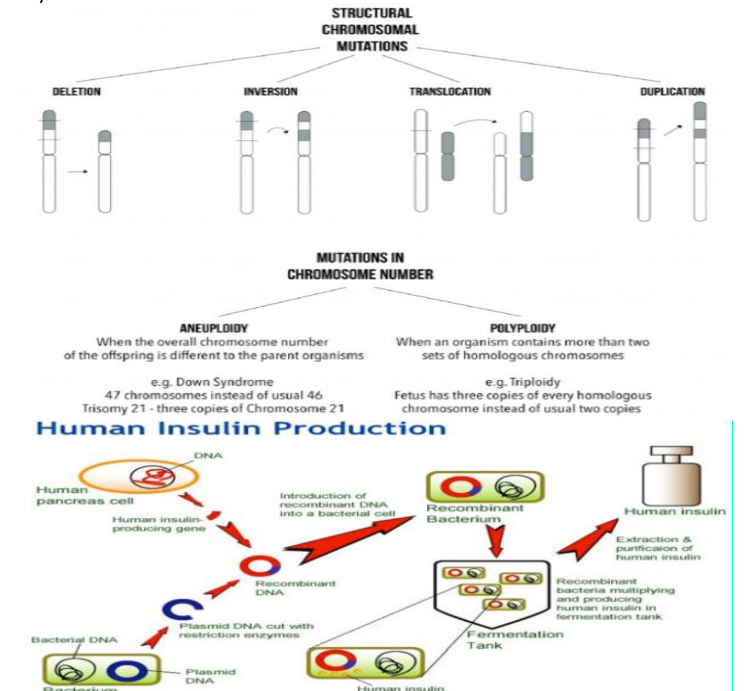
Graphing



6.3.7 Benefits of using genetic technologies-Agricultural- Bt cotton – is a **transgenic** O developed in the mid 1990's and comes from a variety of cotton plant that was subject to pest damage from the larvae of the *Helicoverpa* moth. The moth was controlled using **pesticides**; however, due to environmental damage and adapted resistance, the pesticide needed an alternative. The Bt gene is toxic to the moth and was discovered in the **bacteria Bacillus thuringiensis**. The specific gene was identified, isolated, and inserted into the cotton. Today, India and USA, the two largest producers of cotton, use over 95% Bt cotton. Although there are some economic issues with the use of Bt cotton, it is or has been banned in some regions of these countries. **Medical- Pharming** is the use of genetic engineering to insert genes that code for useful **pharmaceuticals** onto host animals or plants. **Tracy the sheep** (1990-1997) was the first **transgenic** mammal ever created. She was created in Scotland to produce the **human protein alpha 1-antitrypsin** in her milk, a substance regarded in the 1990's as a potential pharmaceutical for the treatments of **cystic fibrosis** and **emphysema**. A deficiency in this protein in humans can produce **lung disease**, and its artificial creation was thought to be a potential success in the disease's treatment. Clinical trials for the engineered protein in 1998 revealed that it developed breathing problems in patients and research for the milk as a remedy for the diseases has not continued since then. **Industry- Biodegradable plastics** – can be broken down and decomposed by living Os, eg: **bacteria** and **fungi**. Analysis of marine animals has revealed a build-up of **microplastics** (**PE-polyethylene**, **PS-polystyrene**, **PP-polypropylene**) in their bodies which affects their health and ability to consume food. **PHB (polyhydroxybutyrate)** is a **polyester polymer** produced by several different micro- Os and is a compostable polymer that can be used to make **biodegradable** plastics. Genetic manipulation has inserted transgenes that code for a particular enzyme which cause certain end products to form such as **PHB**, eg: **cyanobacteria** produce **PHB** in low nitrogen conditions.

6.3.9 Influence of Biotechnology in-Social context–relate to how people live, their needs and requirements. The physical and social setting in which we live. In relation to biotechnology, the biotech techniques that are available to a society are dictated by the specific needs of that society as well as the choices made by government, and the wealth of individuals and the economic status of the country. E.g, **Australia has one of the highest reported cases of melanoma** (type of skin cancer) in the world, up to twice the amount compared to North America countries. **Economic context-(Bt cotton, Golden rice)** are financial factors. The amount of research and development required to bring genetically modified foods to the consumer is huge. E.g, **Philippines is considering one of the 18 countries in the world with richest (highest) biodiversity**. In 2016, pharmaceutical products derived from Philippine's fauna and flora residing in rainforests exceeds \$3.5 billion USD annually. It is a country with the third largest pharmaceutical market in the world. **Cultural Context**–are shared belief systems and characteristics of groups of people. It is dependent on the shared meanings, ideas, beliefs, and characteristics of the people who make up a society. The values, religious and moral beliefs, and educational background of a group of people may influence their opinions and whether they are open to using biotech. E.g, **the Hawaiians has Taro** as a staple food which is also sacred in the Hawaiian culture. In the last century, the S and genetic diversity of taro has greatly reduced due to taro **leaf blight disease caused by a fungus**.

Chromosomal mutations involve the re-arrangement of large portions of DNA, and are impactful upon Os. Serious chromosomal mutations will usually lead to miscarriage early during pregnancy. Chromosomal mutation is usually a result of errors in **meiosis**: when **crossing-over** occurs incorrectly, it may lead to **structural mutations**. When **sister chromatids** are incorrectly separated during **anaphase**, this way lead to errors in chromosome number.



7.1.1 classifying different pathogens that cause disease in plants and animals.

Pathogen-a biological agent that causes disease or illness in a host O. **Vector**-are living Os (usually arthropods) that can transmit infectious pathogens between humans, or from animals to humans, or between plants (mainly viruses, but also bacterial and fungal). **Host**-any O that carries another O (**bacteria**) or agent (**virus**).

Pathogen	Description	example
Prion	Proteinaceous infectious particles. (Not in plant) Also called spongiform. Abnormally folded protein which propagates by transmitting the misfolded protein state to other cellular proteins	Mad cow disease, Kuru, Creutzfeldt Jakob disease
Virus	Non-cellular infective agent consisting of nucleic acid (DNA or RNA) inside a protein coat (envelope). Replicates inside living cells	HIV, Measles, Influenza, TMV
bacteria	Single-celled prokaryotic O (no membrane-bound organelles). Reproduce by binary fission. Secrete toxins, invade cells, and form colonies (biofilms) which disrupts cell function	Salmonella, Tetanus, Fire blight
Protozoa	Single-celled eukaryotic O. Heterotrophic-absorb nutrients from hosts. Secrete toxins, invade cells, and form colonies to disrupt cell and tissue function	Malaria, Dysentery, heart rot disease
Fungi	Eukaryotic, heterotrophic Os with cell walls. Absorb nutrients from environment by secreting digestive enzymes. Reproduce by spreading spores that can release harmful enzymes	Thrush, Ringworms, Tinea, Potato blight
Micro-parasite	Visible with the naked eye. Ectoparasite: lives on an O. Endoparasite: lives in an O. invade and destroy cells, create competition for nutrients	Ticks, Tapeworms, Aphids

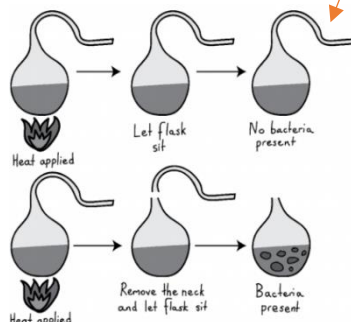
Disease- An abnormal structural or physiological condition that harms the living O and lowering its productivity or usefulness. **Infectious disease**- A disease that is caused by a pathogen or infectious agent, and the disease can be transmitted from one (affected) O to another. **Non-infectious disease**- A disease that is NOT caused by a pathogen and is unable to be passed from one O to another. **Infectious agent**-the pathogen that is capable of infection and causes disease.

7.1.3 design and conduct a practical investigation relating to the microbial testing of water or food samples

Physical	Chemical	Microbial
Turbidity, total dissolve solids Conductivity, pH, hardness Temperature, dissolved oxygen, colour, taste, odour	Free chlorine, Inorganic chemicals (Dissolved salts) Organic compounds	Pathogens Cyanobacteria

7.1.6 Investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including: Koch's postulates

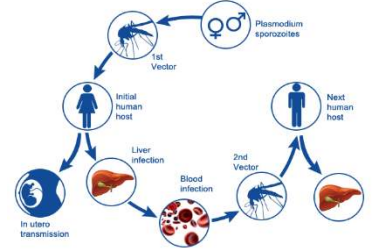
Robert Koch was a German microbiologist working in the late 1800s. He is known as the founder of modern bacteriology, having correctly identified the microbial origins of many diseases, such as anthrax, cholera, and tuberculosis. Most notably, Koch developed a procedure for isolating and identifying disease-causing microbes. This method directly linked microbial growth as a causative agent in disease progression. Koch's postulates: in all Os suffering from disease, the micro-Os must be present in abundance. Micro-Os must be isolated from the diseased O and grown in pure culture. When a healthy O is inoculated with the pure culture, it must develop the same symptoms as the original sick O. Isolate and re-grow the micro-O from newly infected O. If it is identical to the micro-O cultured step 2, it has been identified as the cause of the disease.



7.1.2 investigating the transmission of a disease during an epidemic

An epidemic is a sudden increase in the occurrence of a particular disease among the population of a given area. **Endemics**-The usual level of disease found within a population, where the level of a particular disease is steady and its occurrence is not increasing or decreasing with in a population. **Pandemic**-When a disease reaches above endemic levels on a global scale and epidemics are occurring across multiple countries or continents. **Malaria**

Cause	Plasmodium protozoa
Trans-mission	Vector transmission via Anopheles mosquito: Plasmodium sex cells reproduce in the Anopheles mosquito, forming zygotes in cysts of its stomach wall. Cysts burst and sporozoites travel to the salivary gland of the mosquito, transferring to humans when bitten. Sporozoites travel to the liver, enter RBCs, and continually multiply. Infected cells burst, causing malaria fever. Cycle continues when human host is bitten by mosquito, passing mature cells back the vector. Malaria is prevalent in tropical environments where the climate suits mosquito propagation, and temperature at which plasmodium growth is successful.
Host response	The immune system attempts to clear the body of the plasmodium by cell-mediated response. Anaemia occurs as the RBCs are damaged. This leads to an enlarged liver and spleen, sweating, fever, shivering, and eventual death.
Epidemics	In order to pre-empt epidemics, organisation such as the WHO attempt to forecast outbreaks by assessing a number of factors, including: identification of epidemic-prone areas based on past outbreaks, environmental factors, and climate factors. Vulnerability assessment of population displacement, civil unrest, insecurity of food supply, incidence of other diseases, drug resistance of parasite, and infrastructure (e.g. agricultural projects, dams, flooding). This information is used to stop malaria transmission by proactively breaking the malaria cycle.
Prevention	Mosquito nets, protective clothing, insect repellent
Control	Breaking the plasmodium life cycle by draining swamps and killing mosquitoes.



7.1.4 investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission

Virulence-the ability of the pathogen to infect and cause harm to its host. **Reservoir**-where the pathogen normally lives and reproduces. Can be living Os or the environment. **Portal of exit**-pathway a pathogen uses to exit a host and be transmitted to another. **Transmission**-the way a pathogen spreads disease from one host to the next. **Direct transmission**-include contact BTW the reservoir or infected host and a new host e.g. sexual contact, biting and fihgting. **Indirect-transmission**-requires an intermediate BTW one host and the next. Includes the spread of disease through the air and dust. **Vecotrs**-(mosquitoes, fleas, ticks and flies) spread disease indirectly. Some pathogens require a vector to complete their life cycle and reach maturity in an intermediate host. **Portal of entry**- is often the same as the port of exit from the previous host. The pathogen must be able to access the type of tissues where it can grow and reproduce. **Susceptible host**-an O that will become infected. Depends on genetics, immunity to the pathogen and overall health.

7.1.7 Investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including: Pasteur's experiments on microbial contamination

Louis Pasteur was a French microbiologist who worked during the 1800s. through his experiment, Pasteur disproved the previously accepted theory of spontaneous generation by demonstrating that all micro-Os come from pre-existing micro-Os (germ theory of disease). In his work, he produced the first vaccines for rabies and anthrax, making significant contributions to the field of immunology. He was also involved in the discovery of microbial fermentation and developed the now commonly used technique of pasteurisation. Swan-necked flask experiment- Pasteur's swan-necked flask experiment was used to disprove spontaneous generation. He took flasks with bent necks through which particles in the air could not travel without getting stuck. After filling the flasks, he sterilised the liquid by boiling them at high temperatures. One of the flasks had its neck removed, and one did not. Pasteur showed that only the flask with an open, broken neck was able to grow bacteria in the broth, whereas the other flask remained uncontaminated. Through this he demonstrated that microbial growth was a result of particles in the air and could not arise spontaneously in sterile environments. this finding contributed significantly to our understanding of disease, as the scientific community began to accept that infectious diseases must be s result of micro-Os, originating from some external source.

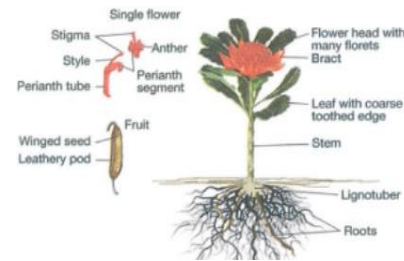
7.1.8 Assess the causes and effects of diseases on agricultural production, including but not limited to: plant diseases-causes-there are a number of infectious agents which contribute to plant diseases, including bacterial, fungal, viral, protozoic, and micro-parasitic agents. In addition, abiotic factors, such as drought, frost, or nutrient deficiency may also have impacts upon plant health. The Australian Government Department of Agriculture and Water Resources (AGDAWR) has identified a number of priority plant pests which greatly affect agriculture in Australia, including: Fruits flies (macro-parasite which infects fruits and vegetables). Wheat stem rust (fungal pest which infects wheat, barley, oats, and rye). Potato cyst nematode (microscopic round worm which eats potato, tomato, and eggplant roots). Sharka (plum pox virus which infects summer fruit, including cherries and plums). **Effects**-infectious diseases affect the ability for plants to carry out normal functions, and therefore have a significant impact on the yield and quality of agricultural products. Due to the reduction in productivity, and costs associated with prevention, plant diseases cost Australia millions of dollars each year. In addition, agriculture is a significant Australian industry, and reduction of its efficiency impacts our ability to trade both locally and internationally. Globally, it is estimated that plant pathogens cause 12.5% of crop losses. Plant disease can also have significant social impacts, particularly in developing countries, by contributing to diminished food security. **E.g.** the Irish Potato Famine (1845-1849) was caused by widespread potato infection, leading to the death of about one million people, as well as mass emigration from Ireland. Plant diseases may adversely affect biodiversity in natural ecosystems, particularly where they have been transported from foreign countries. Biosecurity is the practice of protecting phenomena, such as pests and disease. Australia has strict biosecurity laws and practices in order to maintain our status as a relatively unaffected, isolated nation.

7.1.10 Compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts-appressoria-specialised penetration organs in fungi which are used to infect host plants. **Vectors-(mosquitoes-Malaria)** are a way of overcoming the skin barrier-the use of biting insects. **Penetration** of mucus barrier through physical propulsion or the chemical breakdown of mucus. **Antigenic variation-(Influenza)** change of pathogen cells recognised by a host allowing the pathogen more time to colonise a host before detection. **Biofilm growth**-can cause chronic infection and allows bacteria to increase tolerance to antibiotics and host immune response systems. New spontaneous mutations may occur in the bacterial DNA, making the bacteria resistant to an antibiotic. These bacteria are able to multiply rapidly without competition from sensitive strains and produce cloned offspring that carry the resistance gene. **Host symptoms-(Pneumonic plague)**-such as diarrhoea, coughing and sneezing enhance disease transmission by potentially contaminating, water and food, intermediate surfaces or through direct contact. Some pathogens affect the CNS or **neurochemical communication**, e.g., Toxoplasmosis in rodents reduces their innate fear of cats, increasing the reproductive success of the pathogen.

7.2.2 Investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example: viral pathogens-Banana-Musa S are endemic to Australia, the Malay Peninsula, Indonesia, the Philippines, and New Guinea with the early bananas being small and containing many large seeds and little pulp. In Australia Musa acuminata banksia and Musa jackeyi are Australian native S and can be found growing wild in disturbed rainforest areas of Queensland. The modern banana which was developed in Africa around 650 AD was introduced into Australia in the mid 1800s with plantations started in the Coffs harbour area in the 1890s. Modern cultivated bananas descend from two wild diploid S, M. acuminata (genome A) and M. balbisiana (genome B) with selective breeding choosing seedless plants with fleshy sweet fruit. Since modern bananas are seedless, they are propagated asexually from small suckers taken from the parent plant after fruiting. Modern banana cultivars are polyploids, e.g. AAB, AAAA and ABBB. Bananas are a large herbaceous flowering plant that grow from a corm which is a swollen stem base covered with scale leaves. The plant consists of an apparent trunk that is large leaves rolled over one another. A sucker will shoot from the corm to grow into another banana plant. Inside the rolled part of a leaf there is a bud that will produce flowers which form a large spike that produces the bunch of bananas. Banana bunchy top virus (BBTV) is a single stranded DNA viral disease of bananas that is transmitted from plant to plant by aphids. The virus stunts leaf growth making young leaves appear yellow and 'bunched' and prevents the plant from producing fruit. The virus affects phloem tissue and when the infected cells die, they are lighter in colour causing the area near the leaf stem to have a streaky appearance. The suckers of infected plants will also be diseased. Researchers are investigating the development of resistant varieties of bananas against BBTV, e.g. using biotechnology, genetic engineering and the creation of transgenic banana plants. Most control measures involve controlling the aphid vectors, e.g. using chemical treatments and monitoring alternate vector feeding sites and removing/destroying infected plants. Biosecurity quarantine measures are also used to prevent the transportation of infected fruit and plant material. Banana

7.1.9 Assess the causes and effects of diseases on agricultural production, including but not limited to: animal diseases-causes-as with plant disease, animal diseases may be caused by a number of different infectious agents including bacteria, viruses, protozoa, and macro-parasites. The AGDAWR lists a number of animal diseases which are of particular concern to Australian biosecurity, including: Avian influenza (bird flu-a severe viral disease affecting poultry for which there is no treatment). Foot-and-mouth disease (highly contagious viral infection affecting cloven-hoofed animals, often leading to significant mortality rates in young animals). Bovine Spongiform Encephalopathy (BSE-a fatal neuro-degenerative disease caused by prions, otherwise known as Cow disease). **Effects-economic impacts:** Australia's livestock industry has been fundamental to the growth of the Australian economy in our recent history, contributing around \$ 15 billion in export revenue and \$ 18 billion to GDP in 2017. Animal diseases have the potential to significantly impact this. The CSIRO estimates that a major outbreak of foot-and-mouth disease may cost the Australian economy around \$ 50 billion. **Food security:** animal disease may severely impact agriculture, which has significant impacts upon at-risk populations facing poverty or malnutrition. **Health risks:** animal disease have the potential to infect humans hosts as well. This may affect farmers and handlers, as well as those who consume the products.

7.2.1 Investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example: fungal pathogens-Waratah-(Telopea speciosissima) is a member of the Proteaceae family, has large bright red flowers, is the NSW state emblem and can be seriously affected by fungal infections. T. speciosissima is an erect shrub with alternate leaves and rarely grows taller than two metres in height. The flower structure of the waratah consists of many individual flowers at the end of a branch with a circle of modified leaves called bracts that protect the flower during development. The oldest individual flowers are at the base of the flower head and the youngest individual flowers at the apex. The flower produces large quantities of nectar which attracts a variety of bird and insect pollinators. The nectary is deep at the base of the flower and droplets of nectar leak out to the curled parts of the opened flower and can collect on the red bracts. The perianth which is the outer part of a flower consists of the sepals and petals and forms a tube. As the flower matures the style straightens out and emerges from the tube. The curved shape of the flower means that while honeyeaters are feeding on the nectar with their long-curved beaks, they receive a brush of pollen on the top of their heads from anthers. The seeds are dispersed by birds and mammals though most do not travel far from the parent plant. Waratahs have a lignotuber which means they can resprout after a bushfire has destroyed above ground foliage. Waratahs are difficult to grow from both seedlings and cuttings with many people believing damping off is the main cause for their lack of success in growing waratahs. Damping off refers to fungal infections of the root and crown that affect garden cultivation. To determine if lack of growth and healthiness of a waratah plant is due to the nutrient balance of the soil and/or drainage conditions plant samples need to be checked for the presence of a pathogen. The main fungi pathogens of waratahs are Phytophthora cinnamomic, Cylandrocladium scoparium and Cylandrocarpon destructans. Infected plants have yellow leaves, wilting, dieback and drop and blackening and death of the stem and roots. Phytophthora causes stem rot and Rhizoctonia causes root rot. Anti-rot fungicides help prevent the growth of root fungi. The macadamia nut borer is a serious pest of the waratah.



7.2.3 Analyse responses of animals to pathogens by assessing the physical or chemical changes that occur in the cells and tissues of animals in the presence of pathogens-first line of defence-humans have a number of pre-formed adaptations which inhibit pathogens from entering the body. These includes: **skin**: tightly packed cells forming a protective layer. Pores in the skin secrete anti-microbial fluids, inhibiting surface microbial growth. The outermost layer of the skin (**keratinocytes**) is constantly shed. **Mucous membrane**: cells lining the openings of the body (respiratory tract, urinary, and reproductive systems) secrete a protective layer of **mucous**, which traps pathogens and other foreign particles. **Cilia**: hair-like projections which line the air passages (nose and throat). Movement of these structures pushes pathogens away from the lungs in a wave-like motion. Cilia beat in one direction at about 12 beats per second, working with mucous membranes to move pathogens out of the body. **Chemical barriers**: substances such as stomach acid, alkali conditions in the small intestine, and enzymes in the mouth all act to destroy pathogens. The variation of different pH conditions in the digestive tract ensures that all pathogens are neutralised. Enzymes such as lysozymes dissolves cell membranes to kill pathogens such as bacteria. **Secretions**: fluids are routinely secreted from the sweat glands, hair follicles, and open passages in the body. These secretions contain antimicrobial chemicals which destroy bacteria and fungi, as well as acting to flush out pathogens which may have settled on or in surfaces. **Second line of defence**-apart from pre-formed structures which block entry into the body, animals have also adapted responsive defence systems which fight off foreign Os in the body and stop the progression of disease from pathogens. These are termed the 'second line of defence' as they are activated once a pathogen has passed through external physical barriers and may include: **lymph system**: including the lymphatic vessels, the spleen, thymus, and lymph nodes. It produces WBCs responsible for enacting the immune response. Pathogens are drained to the lymph nodes via lymph fluid, where they can be neutralised or killed by immune cells. **Inflammation**: the dilation of blood vessels and infiltration of inflammatory cells at the site of infection, causing heat, pain, redness, swelling, and acute loss of function. When tissues are damaged, they release histamines, which increase the permeability of proximal blood vessels and allow WBCs to travel more easily to the site. The purpose of inflammation is to eliminate the cause of injury (the pathogen), clear out necrotic cells from the area of infection, and initiate tissue repair by stimulating the flow of blood to the area. By heating the area, pathogens are subjected to

bunchy top disease is an important agricultural disease with outbreaks able to impact on quality production and trade. Biosecurity analysts are computer modelling to see the potential impact of BBTV on large scale banana plantations which consist of Musa clones. This data can be used to construct BBTV exclusion policies and strategies.



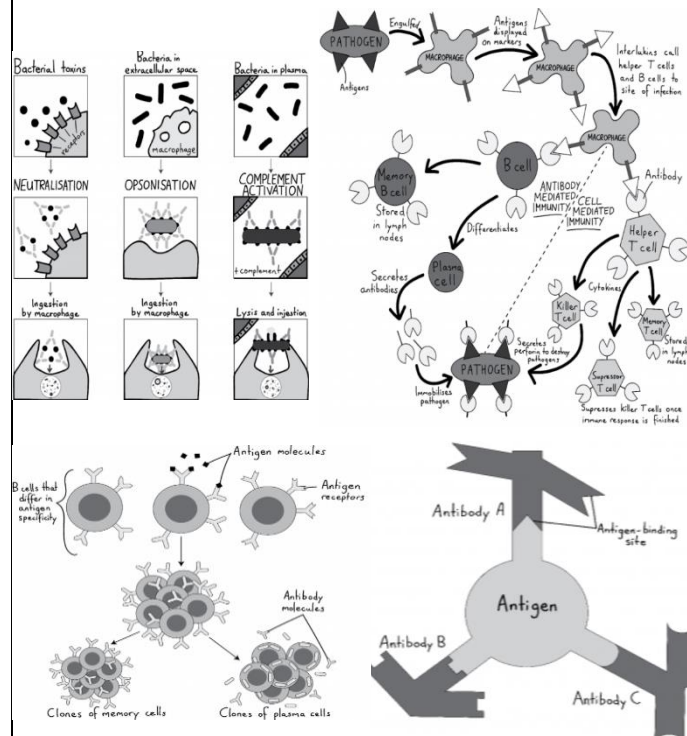
higher temperatures, which may deactivate them, stopping infection. **Phagocytosis:** specialised WBCs, macrophages, and neutrophils can change their shape to engulf pathogens or cellular debris. Once pathogens are enclosed within the immune cells, they can be broken down by enzymes. Phagocytic receptors on the surface of WBCs bind to microbes. Bound materials are internalised, forming phagosomes. The phagosome fuses with lysosomes in the cell, forming phagolysosomes. Microbes inside phagolysosomes are killed/degraded by acidification (pH 3.5-4), antimicrobial proteases and enzymes, reactive oxygen and nitrogen S (which bind to nucleic acids and proteins and oxidises them, causing degradation), and antimicrobial proteins (e.g. defensins). **Cell death to seal off pathogens:** macrophages and lymphocytes may surround a pathogen and undergo programmed cell death (apoptosis). This results in the formation of a cyst, which blocks pathogen movement and any nutrient supply, causing it to also die.

7.3.1 Investigate and model the innate and adaptive immune systems in the human body-

Innate immune system provides non-specific protection against pathogens by responding in a generic manner to all foreign invaders. The innate response is initiated immediately following exposure to a pathogen, or within a few hours-this is a rapid response rate. This stage of the body's defence does not have immunological memory to specific infections, but most aspects are consistently present to provide continued protection to a broad array of pathogens. There are two components of the innate immune response: the first line of defence, and the second line of defence. **Adaptive immune system** is highly specific, providing specialised protection against pathogens which enter the body. This system needs time to develop upon primary exposure to a pathogen (it's not rapid as the innate immune response). However, it has immunological memory, which means response upon secondary exposure and thereafter is stronger and faster. There are two main classes of cells in the adaptive immune system: B cells and T cells. **B cells**-mature in the bone marrow, developing from hematopoietic stem cells. Produce antibodies responsible for antibody-mediated immunity. Once activated by pathogenic antigens, B cells undergo mitosis and differentiate into two types: **Plasma cells** (immediate protection) secrete antibodies of the same antigen specificity as the selected parent B cell. **Memory B cells** (persistent protection): circulate through the body initiating stronger, more rapid responses upon secondary antigen detection. **T cells**-originate in the bone marrow and mature in the thymus. Responsible for cell-mediated immunity. There are a number of different types of T cells, each playing an important role in immunity: **Helper T cells:** assist other WBCs such as B cells in their immunological processes. Once selected by antigens, helper T cells rapidly divide and secrete cytokines (signalling molecules) which help to coordinate the immune response. **Cytotoxic (killer) T cells:** these are the destroyers! These T cells release cytotoxins, such as perforin, and granzymes, that kill target cells by triggering apoptosis. Essentially, they send a signal to an infected cell, forcing it to kill itself and anything inside of it (like a virus). **Regulatory (suppressor) T cells:** control cell-mediated immunity by suppressing the activity of other T cells once the immune reaction has achieved its purpose. Basically, they stop the killer T cells from killing everything. **Memory T cells:** provide immunological memory by remaining in circulation after infection, so that the body may mount a quicker, more effective response upon re-infection. **Antigen-antibody interactions**-antigen are molecular markers present on the surface of all cells. Each type of cell exhibits a certain antigen (all antigens have slightly different shapes) by which they are recognised. The body can recognise foreign pathogens as 'non-self' by their antigens. Antibodies bind specifically to antigens, allowing for a number of immune responses. Antibodies benefit the immune response in three ways: **neutralisation:** antibodies binding to pathogens or toxins to block their effect on host cells. Once immobilised, toxins and pathogens may be degraded, either by macrophages or killer T cells. **Opsonisation:** the binding of antibodies to antigens improves the efficiency of phagocytosis. **Complement system:** by binding to pathogens present in the blood stream, antibodies activate lysis and ingestion of infections. **Clonal selection** is the process by which the adaptive immune system gains its specificity against pathogens. It occurs in both T and B cells, like rapid evolution of our immune system. **Variation:** our body has a pool of variable immune cells-millions of different T and B cells. These naïve cells will circulate in the body until they encounter an antigen. **Selection pressure:** by binding to a specific immune cell receptor, the antigen selects that cell. **Reproduction:** this selected cell rapidly divides, producing lots of copies of the cell which produces antibodies best suited to the antigen. These cells differentiate into the different cells of the immune response (i.e. plasma cells, memory cells). This process is iterative, much like evolution. Within each generation of cell division, there will be small changes to the conformation of antibodies on the cell surface, and this means that antigen affinity may be continually improved.

7.3.2 Explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity-

the immune system enacts a coordinated response to pathogen exposure, mediated predominantly by WBCs, B cells, and T cells. When pathogen first enters the body, it is detected as foreign due to the presence of **non-self-antigens** on its surface. **Inflammation** allows increased blood flow to the site. Increased **permeability** of blood vessels allows WBCs to migrate from the blood into the infected tissue. **Non-specific responses**, including phagocytosis, occur. **Macrophages** engulf pathogens which they encounter and release cytokines to call other immune cells to the site of infection. The macrophages present the foreign antigens on their surface for recognition by B cells and T helper cells which are recruited to the site by **interleukins** (a type of cytokine). **B and T cells** specific to the pathogen are selected by the antigens (clonal selection). B cells differentiate into **plasma cells** and secrete **pathogen specific antibodies** to immobilise the foreign cells. Cytotoxic killer T cells attack pathogenic cells by releasing **cytotoxins** (e.g. perforin). **Memory B and T cells** are produced. Pathogen is cleared from the site. **Suppressor T cells** come in and dampen the immune response, suppressing killer T cells once the infections have passed. Memory B and T cells remain circulating in the blood to provide **long-term immunity**.



7.4.1 Investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease-

Local-immunisation to create herd immunity within local populations. Personal hygiene practices, including washing and drying hands regularly, covering coughs and sneezes, and cleaning surfaces regularly. Safe health practices, including limiting spread of sexually transmitted infections by use of physical contraceptives, and staying at home when you are sick. Provision of public health information to improve public knowledge of diseases and prevention. **Regional**-consideration of **environmental conditions:** **water supply**-access to clean water significantly helps prevent disease. **Sanitation facilities**-inadequate disposal facilities may lead to contamination of water supplies, poor hygiene, and living conditions. **Food**-contamination or poor preservation may lead to a spread of food-borne infections. It is important that food handlers and suppliers are properly trained to limit this. **Climate**-distribution and population size of disease vectors are affected heavily by climate (e.g. mosquitoes which spread malaria breed in warm, humid climates). Flooding-may lead to sewage overflow and water contamination on a large scale. **Improving swift identification:** **continued surveillance**-systematic collection, analysis, interpretation, and dissemination of health data. **Rapid recognition of presence**-disease awareness and reporting from the community, reliant on public health programs and dissemination of information. **Efficient diagnosis of microbial cause**-this is a cornerstone of effective control and prevention efforts, improved by dispersion of technological advances. **Appropriate and efficient responses**, including isolation, treatment, identification of high-risk groups, and the provision of supplies to prevent further transmission. **Global**-communication between countries and with global health organisations is essential (e.g. WHO member states are required to report within 24 hours any disease or event which may constitute a public health emergency of international

7.4.2 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: hygiene practices-

diseases are often caused by transmission of microscopic pathogens. These may be transmitted in food, water, by vectors, in the air, and on the body without our knowledge. To prevent the spread of disease, a number of different procedures may be implemented, ranging from personal to societal measures which we can use to improve public health. **Hygiene practices**-washing hands, cleaning wounds, and undertaking responsible food preparation all minimise the likelihood of micro-organisms entering the body.

concern). Implementation of quarantine measures, which involves things like travel bans into or out of countries significantly affected by disease outbreaks. Monitoring movement of potentially affected individuals. Potential infectious diseases to investigate include influenza epidemics (e.g. avian flu, swine flu), malaria, cholera, yellow fever, and SARS.

7.4.3 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: quarantine-
Quarantine-isolation of an individual for a set period of time in order to prevent the spread of disease. This allows either for the infectious period to elapse, or for symptoms to develop. Quarantine may be used for foods, plants, and animals. By physically detaining diseased individuals, it allows for other individuals in the community to be protected.

7.4.4 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: vaccination, including passive and active immunity-
Vaccination-introducing attenuated pathogenic particles into the body can trigger a small-scale immune response. This allows for specific antibodies to be produced against certain diseases, allowing for a stronger, more rapid response under second exposure. This reduces the likelihood of infection from pathogens such as viruses. E.g. the human papillomavirus (HPV) vaccination program in Australia began for females in 2007 and extended to males In 2013. By 2014, 73% of females and 60% of males turning 15 were fully immunised. There has been a significant fall in HPV-related infection, symptoms, and cervical abnormalities since dissemination of the program across the country

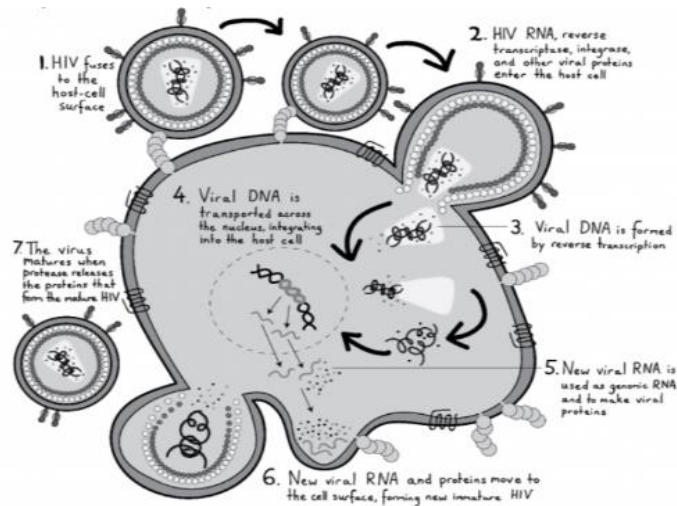
7.4.5 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: public health campaigns-
Public health campaigns: focus on management and prevention and aim to raise awareness and spread understanding about causes and impacts of disease. This may lead to healthier choices in the community, and thus prevent the spread of disease in the population. E.g. the “Ending HIV” initiative to stop the spread of HIV and AIDS was run by ACON- a health promotion organisation funded by the NSW government. ACON aims to provide education, information, and support to at-risk groups. This campaign centred around the slogan “test often, treat earl, stay safe” and was seen in advertisements (billboards, bus stops, posters) and testimonials. ACON also sponsored HIV Testing Week and established an online media presence (www.endinghiv.org.au).



7.4.6 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: use of pesticides-
Pesticides-chemical or biological agents which control pests, including herbicides, insecticides, and antimicrobials. Pesticides are used in agriculture to protect from crop damage, as well as to kill vectors of disease such as mosquitoes.

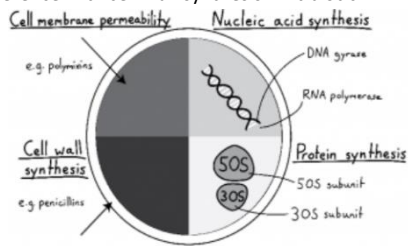
7.4.7 Investigate procedures that can be employed to prevent the spread of disease, including but not limited to: genetic engineering-
Genetic engineering-e.g. engineering malaria-resistant mosquitoes using CRISPR. This involved removing the host factor gene (FREP1) which encodes a protein that helps the malaria parasite survive in the gut of the mosquito and develop for transmission amongst humans is broken. In order to pass the edited trait within the wild population, scientists may be able to use a new technology called **gene drive**: a gene drive improves the odds that a specifically altered gene will be inherited in offspring. To achieve this, the CRISPR-Cas9 system is encoded into the mosquito genome. When passed on to offspring, the guide RNA directs Cas9 to cut the homologous wild-type genome and remove the undesired allele present in the population. The cell then copies the desired, altered gene into the chromosome at the same locus when it repairs the damage. The mosquito now has two identical copies on each chromosome; therefore, all of the offspring will inherit the alteration. The process is repeated each time sexual reproduction occurs, resulting in the spread of the altered trait throughout the population at increased frequency.

7.4.8 Investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example: antivirals-
a class of antimicrobial used to treat viral infections but inhibiting the development of the pathogen inside the host cell. Unlike antibiotics, antivirals are not able to destroy viruses; they only inhibit development of the pathogen. In order to inhibit development, antivirals may target a number of different stages in the virus life cycle. E.g. the viral life cycle of HIV. Types of antivirals include **Before cell entry**-blocking the virus’ ability to infiltrate a cell. This may include drugs which interfere with the virus’ ability to bind to cell surface receptors. Drugs which specifically bind to virus-associated particles, inhibiting their mobility. **During viral synthesis**-drugs which inhibit the ability for a virus to synthesise its requisite components inside the host cell. This may include **inhibition of reverse transcription**-the way that viruses replicate is by inserting their genomic material into the host genome by a process called reverse transcription. Drugs may be developed to stop this process by deactivating the enzymes involved (e.g. **acyclovir for HIV**). **Inhibition of transcription**-after insertion of the viral genetic information into the genome of the host cell, transcription must be performed to express the viral proteins. By blocking transcription factors, this process can be stopped. **Inhibition of protease activity**-viruses include proteases, which are enzymes that cut viral proteins into pieces so that they can assembled to their final configuration. By blocking protease activity, drugs may inhibit the ability for viral proteins to assemble into their final form (e.g. **atazanavir for HIV**). **Release phase**-blocking the ability for a virus to be released from the host cell, inhibiting its ability to transmit to further cells. By blocking molecules found on the surface of viruses, drugs prevent the release of viral particles (e.g. **zanamivir for influenza**). It is very difficult to develop effective antiviral drugs because viruses use host cell machinery in order to replicate. The key is to target molecules which interfere with the virus, but not with host cell processes. which exists between different viruses. Furthermore, continued use of antivirals may result in antiviral resistance where the effectiveness of drugs becomes reduced overtime. Research and development of antivirals is also very expensive, resulting in high prices of treatment, e.g. a twelve week supply of hepatitis C drug costs to \$ 113, 400. Such exorbitant costs may also be due to monopolies in the pharmaceutical market, and so companies are able to drive up prices to increase profit.

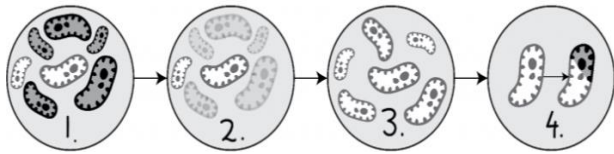


7.4.9 Investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example: antibiotics—a class of antimicrobials used to treat bacterial infections.

This may be achieved either by killing the infective bacteria or inhibiting its growth. There are many ways by which drugs may stop the growth of or kill bacteria. This involves inhibition of a number of key processes in bacterial growth including interference with cell membrane permeability. Interference with nucleic acid synthesis. Interference with protein synthesis. Interference with cell wall synthesis. Antibiotic resistance is an increasing issue for disease control. There are a number of human factors that have contributed to this issue, including misuse and overuse of antibiotics. This leads to the evolution



of bacterial strains which do not respond to antibiotic treatment, as they have developed mechanisms to either avoid or deactivate drugs. Antibiotic-resistance strains of bacteria evolved using the fundamental principles of evolution by natural selection. Within every population of bacteria, a few are antibiotic-resistant due to natural variation and mutation. Whenever someone uses antibiotics, most of the bacteria are killed, but the few resistant ones survive. These are then able to reproduce, with antibiotic resistance becoming the dominant trait in the population, eventually leading to a new 'superbug' strain. Bacteria are able to pass genetic information to each other using plasmids. This process speeds up the evolution of species, as those antibiotic-resistant individuals can pass their genes for resistance onto others.



7.4.11 Interpret data relating to the incidence and prevalence of infectious disease in populations, for example: mobility of individuals and the portion that are immune or immunised—**incidence**—the frequency of new cases of a disease over a specified period of time. **Prevalence**—the proportion of a particular population affected by a disease. **Mortality**—the number of deaths within a particular population as a result of a certain disease, over a specified period of time.

7.4.10 Investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic—**Swine flu**—the emergence of a novel influenza virus was first reported in early 2009 in Mexico. This was identified as H1N1 virus, commonly termed 'swine flu'. In April 2009, the WHO defined the epidemic as a 'public health emergency of international concern' after more than 882 reported cases in Mexico and several in the US, with 62 deaths. By June 2009, the WHO declared the outbreak as a pandemic, with more than 30,000 reported cases across 74 countries. By 2011, approximately 1.5 million people were infected in 214 countries, including 25,000 confirmed deaths. Particular groups, including pregnant women, Indigenous peoples, and the morbidly obese, were identified as facing greater risk of serious illness from the infection. **Management and quarantine methods**—in order to control the spread of the infection in Australia, significant public health action was required. The Australia Health Management Plan for Pandemic Influenza (AHMPPI) outlined the following phases for pandemic response: **Delay (April 2009)**—the objective of this stage was to prevent or slow entry of the virus into Australia. This was enacted by increasing border control measures and increasing vigilance. Thermal imaging was used to screen arrivals to all international Australian airports, in addition to arrivals cards. Customs officers also surveyed aeroplane cabins for anyone exhibiting flu symptoms prior to disembarkation. **Contain (May 2009)**: after identification of numerous cases within Australia, contain phase was activated. These measures aimed to prevent community transmission. All states were authorised with the option to close schools if they identified that students were at risk. Students returning from overseas travel in widely affected countries were told not to return to school for a week after re-entering Australia. A Commonwealth hotline for swine influenza was set up, so that suspected cases could be reported quickly. Public health posters with hygiene practice recommendations (particularly for health professionals) were also published, advising on the use of protective equipment, hand washing, and sterilisation. **Protect (June 2009)**: in recognition that efforts to contain swine flu had not been entirely effective, this phase identified high-risk groups, and aimed to protect those most at risk of developing severe illness from infection. A large-scale public vaccination was rolled out across Australia in September 2009. **Evaluation—preparedness**: previous outbreaks of other influenza strains in the Asia-Pacific region in 2003 and 2004 meant that Australia had instituted comprehensive pre-pandemic planning before the appearance of the H1N1 virus. This involved designing mechanisms for forward planning and forecasting, communication, surveillance, reducing transmission, and optimising health services. Responses to pandemics based on these plans were trialled in large-scale pandemic exercises (2006 and 2008). **Public health leadership**: a significant number of personnel were involved in the pandemic control, with teams tracking patients, tracing contacts, running laboratory tests, collecting and analysing data, as well as creating supply chains for medications and vaccines. Coordination of these efforts was attributed to the successful leadership of the Minister for Health and Ageing and the Commonwealth Chief Medical Officer. **Media management**: frequent comparisons were made between the outbreak and the 1918 flu pandemic, which was a significantly more serious epidemic infecting 500 million people worldwide. This hype of public fear meant that emergency departments were overwhelmed, and resources such as antiviral drugs were drained. This practice jeopardised access to resources for higher risk patients.

7.4.12 Interpret data relating to the incidence and prevalence of infectious disease in populations, for example: Malaria or Dengue Fever in South East Asia—**Dengue fever**

Description	Mosquito-borne tropical disease caused by dengue virus. Spread by the Aedes type of mosquito. Infection results in fever, headache, joint pain, and rashes. may lead to development of dengue haemorrhagic fever (DHF—bleeding, low levels of blood platelets, blood plasma leakage), or develop into dengue shock syndrome (dangerously low blood pressure).
Incidence	390 million infections globally per year (96 million manifest clinically). Current global estimates are that 3.9 billion people in 128 countries are at risk of infection. 75% of global population exposed to DF live in the Asia-Pacific region.
Prevalence	Found in tropical and sub-tropical climates. Most countries in south-east Asia experience a higher burden of DF or DHF with frequent cyclical epidemics (3–5-year cycles). Higher ratio of males than females hospitalised. Typically affects children (2–15 years old) at a high rate than adults. Epicentres of outbreaks are located in major cities, mostly affecting urban and semi-urban areas. Currently associated with rainy season and El Nino phenomenon. Expected rate of DF will increase over time due to viral evolution, climate changes, globalisation, travel and trade factors, and settlement and socioeconomic factors.
Mortality rates	22 million deaths globally per year.
Sources	World Health Organisation DengueNet database: www.who.in/denguecontrol/en/ . Rajesh Bhatia, Aditya P Dash, Temmy Sunyoto, 'changing epidemiology of dengue in South-East Asia', South-East Asia Journal of Public Health (2013). Natasha Murray, Mikkel Quam, and Annelies Wilder-Smith, 'Clinical Epidemiology of dengue: past, present, and future prospects', Clinical Epidemiology (2013). End-Eong Ooi and Duane J. Gubler, 'Dengue in Southeast Asia: epidemiological characteristics and strategic challenges in disease prevention', Caderos de Saude Publica (2009).

7.4.13 Evaluate historical, culturally diverse and current strategies to predict and control the spread of disease - the ability to predict and control the spread of disease has historically been linked to our ability to collect data on patterns of disease through surveillance and analyse this data to ascertain trends and patterns. The development of the field of epidemiology has therefore been crucial in improvement of public health systems worldwide, allowing professionals to identify causes, risk factors, and in turn propose mechanisms for control of diseases.

460 BCE-370 BCE	Hippocrates, an ancient Greek physician, sparks the idea of collecting and analysing data to predict and control disease. He believed that disease was a result of local conditions and collected data about the natural environment to determine when and where illnesses would occur.
1348 CE	Venetian Republic attempts to control the spread of the bubonic plague by appointing public health officials to monitor incoming ships and exclude those with infected individuals aboard.
1377 CE	The city of Marseille uses quarantine to control the spread of disease by detaining individuals who had travelled from plague-infected areas for 40 days.
1662 CE	John Graunt publishes 'Natural and Political Observations Made upon the Bills of Mortality, which quantifies the patterns of disease within the London population in order to study the cause of disease.
1741 CE	First example of legislative surveillance of disease-Rhode Island passes a law requiring businesses to report instances of contagious disease among patrons.
1776 CE	Johann Peter Frank formulates a comprehensive health policy for Germany, detailing injury prevention, maternal and child health, and public water and sewage treatment. This had impacts in other European countries such as Hungary, Italy, and Russia.
1834 CE	Edwin Chadwick identifies the link between poverty and disease using surveillance data in England. This led to reform in the Poor Law system to improve disease control and spread in the population.
1849 CE	Dr. John Snow investigates the causes of cholera in the 19 th century. By mapping cases on a dot-map, Snow illustrated a cluster of affected individuals around a public water pump, identifying it as the source of the outbreak.
1965 CE	Director General of the WHO establishes the epidemiological surveillance unit to predict and track the spread of communicable disease globally.

Current strategies for prediction

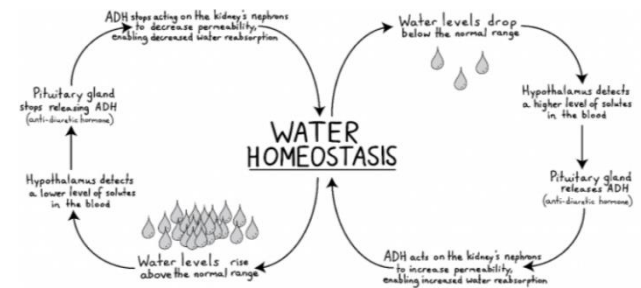
Event-based surveillance: based on reporting by healthcare professionals, as well as news reports, social media, and internet searches. **Global Public Health Intelligence Network** (GPHIN) systematically scans a multitude of informal sources (web sites, electronic discussion forums, new reports, online newspaper) for unusual disease events and rumours for outbreaks. Data is investigated, then verified through official sources. GPHIN issued the first alert of unusual respiratory illness for the SARS outbreak in 2002, triggering an international response from the WHO. **Event-based surveillance** is particularly effective in countries where there is weak or minimal public health infrastructure and can provide real-time data on local disease activity. **Web-based surveillance:** e.g., Google Flu Trends monitors real-time influenza's activity based on the finding that there is a correlation between the number of people searching the web using influenza's-related keywords and the number of people actually experiencing influenza symptoms. It has been shown that this tool is able to predict region outbreaks 1-2 weeks earlier than disease control centres. **Modelling of disease emergence and spread:** A model uses prescribed rules to describe how an infectious disease may spread within a population. This requires collection of data on factors such as: **Human factors:** population density, travel and trade, and how these each affect disease transmission. **Ecological factors:** climate change, agriculture practices, and how these each affect disease transmission. Using models, scientists may conduct computer simulations of outbreak events. This information may be used to inform pre-emptive public health policies and to design rapid response plans to epidemics or pandemics.

7.4.14 Investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example: bush medicine - is a term describing the skills and practices used to maintain health, based on Indigenous beliefs and experiences. This refers not only use of native flora and fauna traditionally prepared, but includes preventative and diagnostic techniques, and treatment of mental illnesses. Bush medicine denotes a holistic view of health, emphasising the interplay between physical, emotional, social, and spiritual aspects of wellbeing. Indigenous Australians are a diverse people, composed of many culturally distinct groups, each with their own individual practices. Thus, there is no single set of Aboriginal medicine and remedies. The National Aboriginal Health Strategy has defined bush medicine "not just physical wellbeing of the individual, but the social, emotional, and cultural wellbeing of the whole community. This is the whole-of-life view, and it also includes the cyclical concept of life-death-life". Bush medicine includes the use of the plant materials, such as bark, leaves, seeds, and some animal products, to create herbal medicines. **Examples: Tea tree oil** (Melaleuca alternifolia)-Bundjalung Aboriginal peoples from the NSW coast. Crushed tea-tree leaves applied as a paste to wounds. Brewing into a tea for throat ailments. Strong antiseptic. Used in western medicine to treat fungal infections and acne. **Eucalyptus oil** (Eucalyptus sp.): infusions used to treat muscle aches, fevers, and chills. Western uses commercially in mouthwash and cough lozies. **Kakadu plum** (Terminalia ferdinandiana): extremely rich source of vitamin C (50 x more than oranges). Found in Northern Territory and Western Australia woodlands. Major food source for tribes in the area. **Desert mushrooms** (Pycnoporus sp.): bright orange mushroom variety. Used to treat sore mouths and lips, or babies with oral thrush, by sucking on the fruiting body of the plant. **Emu bush** (Eremophila sp.): used by tribes in the Northern Territory to treat sores and cuts. Current research has identified the leaves of the plant as possessing strong antibiotic properties, equal to established pharmaceuticals. **Witchetty grub** (Endoxyla leucomochla): made into a paste, these insects were used to treat burns and soothe skin. Recorded use by Central Australian tribes. **Kangaroos' apple** (Solanum laciniatum): Juice from the fruit was applied to swollen joints to relieve pain. Research has found that the fruit contains a steroid which promotes the production of cortisone, accounting for its healing properties.

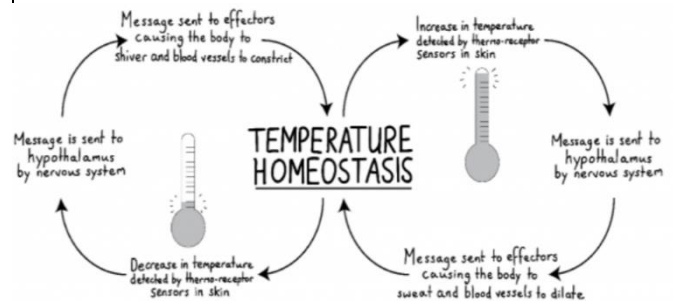
7.4.15 Investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example: smoke bush in Western Australia

Conospermum, or **smokebush**, is a plant native to the south-west region of Western Australia. Smokebush has been used traditionally by the Aboriginal people of the region as a medicine. In the 1960s, the US National Cancer Institute was granted a licence by the Western Australian Government to collect plant samples to screen them for the presence of cancer-fighting molecules. Specimens of the smokebush plant were tested but found to be ineffective in treating cancer. However, in a quest to find potential treatments for the growing AIDS epidemic in the 1980s, the smokebush was again screened, and miraculously found to be one of only four plants out of 7000 which contained an active molecule, **conocuvone**, able to combat the HIV virus in low concentrations. the US Department of Health and Human services filled patents (US in 1993 and Australia in 1994) for exclusive rights to use the compound in AIDS treatment. Rights to develop the patent in Australia were licensed exclusively to a Victorian Pharmaceutical company, AMRAD. To gain access to rights over the plant for research, \$1.65 million was paid to the WA Government. Additionally, estimations were made that if **conocuvone** was successfully commercialised, the WA government would likely recoup royalties of up to \$100 million per year. Whilst this may at its face seem like a win for the WA Government, the smokebush story is another in a long line of incidents in Australian history where the significant contributions of our Indigenous peoples have been forgotten or disregarded, and their expertise and importance to Australian culture has been exploited for financial gain. Inter-generational nurturing of the plant, and its continued use in Indigenous medicine resulted in the knowledge that the smokebush is a healing plant, indicating the potency which led to its selection as a plant to be screened. However, no royalties, compensation, or even acknowledgement was ever presented to the Aboriginal Australians of Western Australia. This demonstrates a significant flaw in our patent law system and its inability to protect traditional knowledge of Indigenous peoples. For a system of law which has been designed to protect an individual's right to their intellectual invention, their property, this must be viewed an egregious breach. This also represents a threat to Aboriginal communities and their traditional cultural practices. There is a possibility that the rights to use an entire species of flora may be sold to large multi-national drug companies. This would prevent groups, including Aboriginal peoples, from using such plants subject to exclusive agreement (presumably involving a paid licence). Essentially, the patenting of traditional medicinal plants may prevent Indigenous Australians from continuing to autonomously use their own cultural knowledge. There is a known wealth of information which Aboriginal and Torres Strait Islander Australians possess, stemming from a continued connection to the land for over 65 000 years, and a region-specific knowledge of not only resources, but also of horticulture and how best to prepare medicines using different parts of plants. It is important that when undergoing bioprospecting practices, companies consider the contribution of Indigenous peoples to discovery, and the rights which may accumulate because of cultural practices. It is also extremely important that we understand the culture-specific rules associated with Indigenous knowledge. Ownership manifests itself in very different ways across Indigenous and Western societies. As Aboriginal Australians have used oral histories to pass their cultural information through the generations, there may be complex rules governing the dissemination of information. Some information is sacred, only able to be used by those within a group possessing certain authority, and not permitted to be made public. Customary laws and community values should be respected during any commercialisation process, so that Indigenous knowledge may be protected as intended. This is particularly important considering the limited control our Indigenous peoples now have over their homelands. **Positive developments**—there currently exist several patents which have been successfully designed with Indigenous co-owners, where business and entrepreneurship have been balanced with cultural necessities. The following is an example of how research may be responsibly enacted to include Indigenous peoples, and respectfully lead to the mutual success of stakeholders in culturally significant information. **E.g., The mudjala plant patent**—the myardoo majala tree has been known to the Nyinkina Mangala community as possessing healing powers and pain relief. It also features in their creation story of the Fitzroy River. Elders of the community approached researchers at Griffith University to create a research partnership. This resulted in identification and isolation of the active compounds in the plant which contributed to its medicinal properties. The Jarlmandang community and Griffith University became joint patent holders of the biotechnology. It is hoped that in the future, the IP technology will be successfully commercialised. The Jarlmandang Buru community continues to actively participate in harvesting and monitoring trials which will ensure that many Aboriginal communities benefit from any large-scale commercialisation opportunity.

8.1.1 construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to:

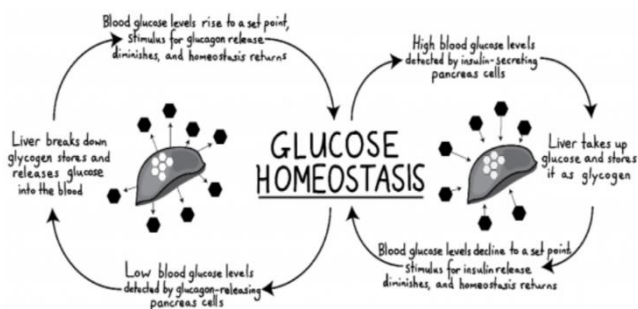


temperature homeostasis is the process by which an O maintains a stable



internal environment, despite fluctuating external environmental conditions. It is essential for Os to maintain a consistent internal environment for several reasons. Enzymes are very important biological catalysts, which require specific conditions to function at their best. By maintaining constant conditions (temperature, pH, levels of water, etc.) in the body, our enzymes can carry out their functions efficiently. Secondly, cells survive best in an isotonic solution, which needs to be maintained so that cells do not shrink or expand and can efficiently carry out their individual functions. Homeostasis occurs in two stages; firstly, the body detects changes from the stable state (the body at optimal conditions), and secondly, the body counteracts the changes. To demonstrate the process of homeostasis, we can construct negative feedback loops, which show how the body detects, processes, and counteracts changes in the external environment.

8.1.1 construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to: glucose



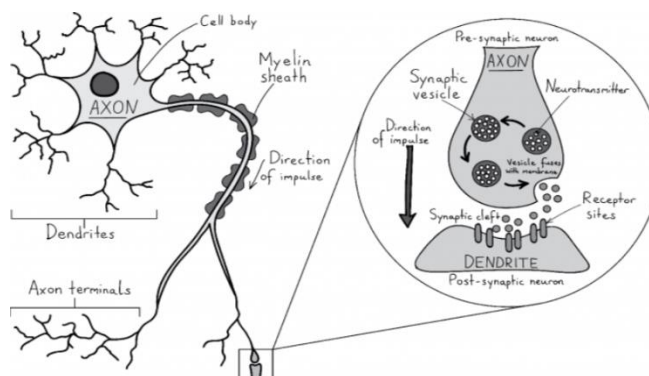
8.1.2 investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including: trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis-endotherms-Os which are able to maintain a constant internal body temperature, independent of the environment.

Adaptions	Trends and patterns
Behavioural	Movement -shaded areas or wet environments help regulate heat exposure. Deliberate use of muscles -muscle contractions produces metabolic heat. Sunbaking -increasing surface area available for heat absorption. Licking -enabling more heat to be evaporated through saliva. Drinking water -to replenish fluids to maintain cells in a consistent, isotonic state.
Structural	Insulation (e.g., feathers, hair, fur) trap a layer of air next to the skin which reduces transfer of heat to the environment. Surface area to volume ratio -as more compact bodies reduce the surface area available for heat exchange, allowing for animals to retain heat more effectively.
Physiological	Vasoconstriction -(narrowing the blood vessels) and vasodilation (widening blood vessels) allows animals to regulate the surface area to volume ratio of their circulatory systems to retain or expel heat when required. Metabolic rates -can be increased to increase the production of heat energy internally or decreased to cool body temperature. Muscle contraction -(shivering) or making small movements in the skeletal muscles to produce heat energy. Sweating -as perspiration allows sweat to evaporate from the surface of the skin, which has an evaporative cooling effect. Panting -allows evaporation from internal body surfaces, such as nasal passages, mouth, and lungs which also has an evaporative cooling effect

8.1.2 investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including: internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways-**hormones**-are signalling molecules used by the body to regulate physiology and behaviour. They are produced in the glands and transported around the body using the circulatory or lymph systems. Hormones affects cells by binding to specific receptors on their surfaces, enacting changes in a process called signal transduction.

Step 1: reception-the hormone binds to a receptor displayed on outside of the cell. **Step 2: transduction**-the binding event on the cell surface triggers a signalling cascade within the cytoplasm. **Step 3: response**-cell signalling initiates a response, usually in the form of gene transcription withing the nucleus. Through these steps, we can see how hormones are able to enact physiological changes by regulating the expression of certain proteins in the cell to dictate cell structure and function. Hormones may affect the metabolism of target cells, tissues, and organs, by either increasing or decreasing levels of activity. Hormones are also responsible for regulating digestion, respiration, sleep, growth, and development, reproduction, and mood. **Neural pathways**-occur as part of the **nervous system**: **Central Nervous System (CNS)**: brain and spinal cord-control centre that coordinates responses within the body. Receives information from the PNS, interprets the information, and then initiates an appropriate response by sending messages to the PNS to enact effector response.

Peripheral Nervous System (PNS): nerves branching from the CNS throughout the body-facilitates communication in the body. Passes messages to and from the CNS rapidly, allowing the body to respond to changes (both internal and external). The nervous system is composed of neural cell (neurons), which pass information in the form of chemical signals around the body. The movement of nerve impulses is facilitated by the transmission of an action potential along the nerve axon, which triggers the release of neurotransmitters, and therefore creates signal transduction between cells. Neurons interface with cells, allowing them to communicate through release of neurotransmitters. These chemicals trigger signalling cascades within cells, enacting specific physiological responses. Neural pathways are excited by the triggering of sensory receptors (thermoreceptors detect changes in temperature, mechanoreceptors detect changes in pressure or distortion, and chemoreceptors detect changes in levels of certain chemicals). When these receptors are triggered, they pass the signal onto neurons, which then pass the signal onto effectors. Effectors work to counteract the detected change, thus maintaining homeostasis within the O. Stimulus → Receptor → Control Centre → Effector → Response.



8.1.2 investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including: mechanisms in plants that allow water balance to be maintained-plants may be adapted in a number of ways which allow them to balance water levels. Mechanisms for water retention are observed widely in Australia flora, as they are often subject to harsh, drought-like conditions. **Theory of Transpiration-Cohesion-Tension theory**: the mechanism by which water flows through the xylem of plants is due to the combined effects of: **Transpiration** (evaporation of water through the stomata of plants).

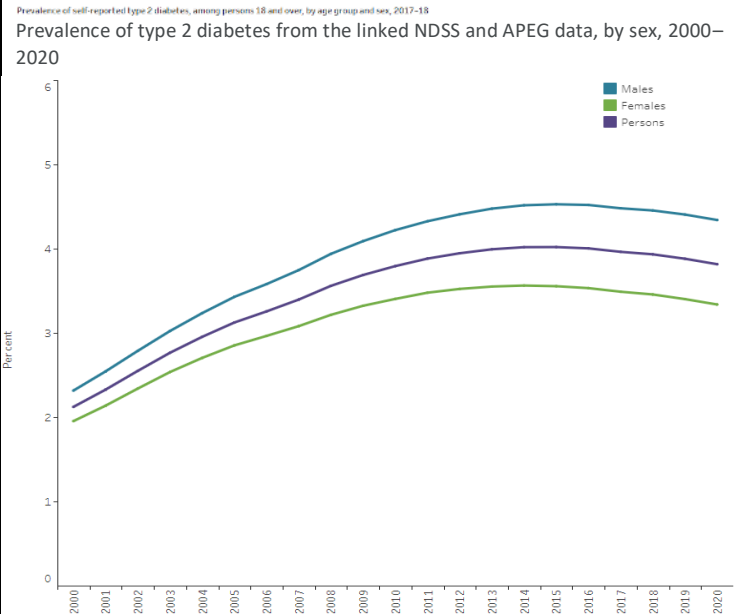
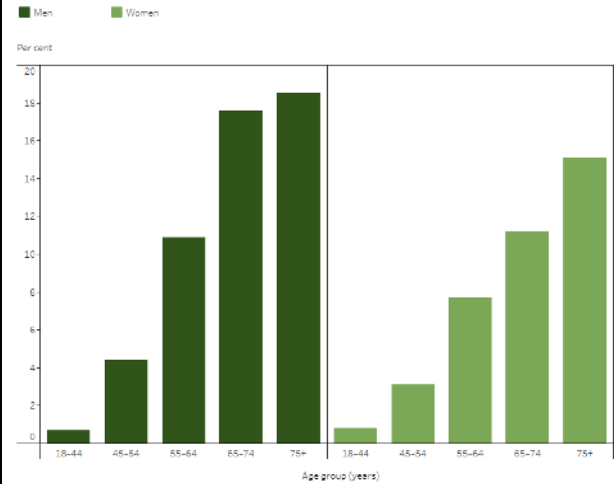
Cohesion (that water molecules are attracted to each other, so will move in a cohesive stream). **Tension** (water molecules are attracted to the surfaces which they touch). By regulating the levels at which transpiration occurs, plants can retain, or release water as required. This may be achieved by: **smaller leaves**-reduces surface area to volume ratio (reducing surface available for transpiration reduces water loss). **Closing stomates**-inhibits ability for water to leave plant through the leaves. **Movement**-angling leaves away from the sun at different times of the day may reduce rates of transpiration (cooler temperatures result in less evaporation). **Dropping leaves during summer and droughts**-conserves water to essential parts of plant. **Large cavities** for water storage in stems/trunks.

8.2.1 investigate the causes and host responses of non-infectious diseases in humans, including but not limited to: genetic diseases-Gene or chromosomal abnormalities caused by point or chromosomal mutations. Such mutations may result from errors during gamete formation, or exposure to mutagens. Genetic diseases may be inherited from parents, or as a result of acquired changes to pre-existing genes. E.g., **Down syndrome** (chromosomal abnormality), **cystic fibrosis** (single gene disorder).

8.2.1 investigate the causes and host responses of non-infectious diseases in humans, including but not limited to: diseases caused by environmental exposure-Interaction with the environment and exposure to physical factors such as radiation. Exposure to harmful or toxic chemicals such as toxic metals or noxious gases. E.g., **Minamata** (ingestion of large amounts of mercury). **Mesothelioma** (cancer as a result of asbestos exposure).

8.2.1 investigate the causes and host responses of non-infectious diseases in humans, including but not limited to: nutritional diseases - Issues with diet (e.g., excessive or insufficient consumption of food). Problems with digestion. Consumption of incorrect amounts of specific foods (e.g., essential vitamins, amino acids, fatty acids or minerals). E.g., Scurvy, Rickets, coronary heart disease, type 2 diabetes. Scurvy-a lack of vitamin C in the diet. Rickets-lack of vitamin D.	8.2.1 investigate the causes and host responses of non-infectious diseases in humans, including but not limited to: cancer - Many cancers are caused by many different factors, including infectious agents, genetic disorders, exposure to mutagenic factors in the environment (e.g., radiation), and lifestyle habits (e.g., smoking, alcoholism). E.g., Cervical cancer (from HPV exposure). Breast cancer (from inheritance of BRCA1 gene). Melanoma (from repeated sun exposure). Lung cancer (from smoking). Tumour is abnormal growth of cells. PTEN is a tumour suppressor gene that limits the amount of cell division and encourages cell death. Mutation of BRAC 1.																												
8.3.1 collect data that shows the incidence, prevalence and mortality rates of non-infectious disease, for example: nutritional diseases-type 2 diabetes <table border="1"> <tr> <td>Description</td><td>Condition in which the body becomes resistant to insulin and is gradually unable to produce it. This leads to a build-up of glucose in the blood, which can cause damage to systems in the body.</td></tr> <tr> <td>Causes</td><td>genetic factors, sedentary lifestyles, and unhealthy diets</td></tr> <tr> <td>Symptoms</td><td>Frequent urination, Increased thirst, Fatigue, tiredness, Sudden weight loss, Itching around the genitals, Slow healing cuts or wounds, Blurred vision</td></tr> <tr> <td>Host response</td><td>The body may make enough insulin early in the disease but does not respond to it effectively. As type 2 diabetes progresses, the pancreas gradually loses the ability to produce enough insulin. Stem cells stops converting into the hormone insulin.</td></tr> <tr> <td>Prevention</td><td>Maintaining a normal weight, eating a healthy and varied diet, and exercising regularly</td></tr> <tr> <td>Treatment</td><td>Lifestyle changes: eating well-helps to manages levels of blood glucose. Diet should include high-fibre, low-fat foods (fruits, vegetables, grains) and avoid refined sugars and carbohydrates. Low glycaemic index foods are also recommended as they help to maintain more stable blood sugar levels. Exercising-lowers blood pressure and helps insulin to work effectively, reducing the risk of heart disease. Physical activity also helps to lower blood sugar levels, as glucose is being used by cells to enact aerobic respiration. Monitoring: testing blood glucose levels regularly allows patients to effectively manages their treatment, ensuring that these levels stay within a healthy range. Treatment: the first step in diabetes treatment is usually e medication called metformin which decreases the amount of glucose produced in the liver, skeletal muscle, and adipose tissue. Overall. This helps to decreases blood sugar levels. as type 2 diabetes progresses, and the pancreas continues to become less effective at producing insulin, patients may also be required to take insulin injections. Insulin signals to cells to take up glucose, and therefore helps to lower blood glucose levels in diabetics. For obese patients suffering from diabetes, bariatric surgery may also help in reducing its effects. This involves reducing the size of the stomach, either by removing a portion of the stomach, or restricting it using a gastric band. This may potentially reset metabolism, essentially curing diabetes.</td></tr> <tr> <td>Future direction</td><td>An increased understanding of the factors contributing to development of type 2 diabetes would help scientists and public health workers to design better prevention measures and create effective treatment and management strategies. Future research may be conducted into understanding genetic risks and analysing genomes to identify genes which may place an individual at risk of developing the disease. This would allow the tailoring of pre-emptive, personalised medicine, which would be particularly effective as prevention is the best form of treatment. Type 2 diabetes is commonly treated by taking insulin shots. Synthetic production of insulin is an important area of research, particularly in terms of how to make this process faster, more efficient, and less expensive, in order to increase worldwide access. This could involve genetic engineering, cloning, or large-scale synthetic protein production. An example of current research in this area is the 2017 Sydney University IGEM team, who aimed to provide an open-source solution to the issue of insulin access globally. Another treatment option for diabetes is whole-organ pancreas transplantation, a surgical procedure in which an entirely new, functional pancreas is placed into a patient in order to cure the disease. This option may be developed by research into creation of artificial pancreases to substitute function. This may be facilitated by the development of 3D printing techniques, or growth of pancreatic cells in vitro. Instead of inserting entirely new organs into patients, islet cell transplants may be an option for diabetics. As islet cells are dysfunctional in diabetic patients, unable to properly produce insulin, transplanting these specific cells into patients may represent as a less complex and less invasive option. 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Research would need to be done into how to grow healthy cells outside the body, and how to insert these into the pancreas in order to restore function	8.3.1 collect data that shows the incidence, prevalence and mortality rates of non-infectious disease, for example: diseases caused by environmental exposure-melanoma, breast cancer <table border="1"> <tr> <td>Description</td><td>a specific kind of skin cancer. It begins in skin cells called melanocytes. Melanocytes produce melanin, the substance that gives your skin colour.</td></tr> <tr> <td>Causes</td><td>The exact cause is unknown but environmental and genetic factors play a role. Risk factors include Exposure to UV (ultraviolet) radiation. Gene mutation. Fair skin. Family history increases the risk. Weakened immune system</td></tr> <tr> <td>Symptoms</td><td>The symptoms include: The earliest symptom is typically a change in an existing mole or a new mole arising. The development of a new pigmented or unusual-looking growth on your skin The ABCDE's of melanoma helps identify the disease condition Asymmetry: The mole may not be uniform; one half looks different from the other half. Border irregularity: The moles may be uneven or notched. Colour: The moles may be of different colours and irregular patterns. Diameter: The mole may be bigger in size, at least a 6 mm size. Evolving: The mole may be changing colours, size and texture and might also bleed</td></tr> <tr> <td>Host response</td><td>DNA repairs will try to repair the damaged DNA. Antibodies are produced to fight the melanoma cells, but they cannot cope for long time.</td></tr> <tr> <td>Prevention</td><td></td></tr> <tr> <td>Treatment</td><td>Surgery involves removing the tumour and the skin around the tumour to ensure that none of cancerous cells are left behind, to reduce the risk of a recurrence of the melanoma. Advanced melanomas are treated by radiation, chemotherapy, targeted therapies, and immunotherapy. The type of treatment used will depend on numerous factors, including how far the melanoma has progressed, and the person's age and health. Radiation is when cells are exposed to X-rays, normal cells can repair their DNA more easily than cancer cells can. X-rays damage the DNA of the cancer cells and kill them. Care must be taken in directing the radiation to the specific cells that are intended for destruction. This radiation can be administered externally or by placing a radioactive source in the body near the cancer cells. Chemotherapy drugs slow the growth of cancer cells, but they have not been particularly effective in treating melanoma, and so are not used to treat this form of cancer as much as in other cancers. Targeted therapies involve the use of drugs that affect the molecules that control that control the growth of the tumour cells, which effectively stops the growth and spread of the tumour to other organs. Melanoma cells have mutations that cause the uncontrolled division of cells, leading to the growth of tumours. Targeted therapy drugs interrupt the pathways that cause this uncontrolled division of melanoma cells. Immunotherapy treatment effectively cause the body's own immune system to fight the melanoma. Two approaches are in the early stages of development and use. One of the approaches is to use drugs called checkpoint inhibitors that cause the immune system to recognise and destroy melanoma cells. Cancer cells cause the immune system to ignore them. Checkpoint inhibitors reverse this effect and have proved successful in a number of patients. The other approach is to use vaccinations as a method of treatment rather than prevention. 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Host response	The body may make enough insulin early in the disease but does not respond to it effectively. As type 2 diabetes progresses, the pancreas gradually loses the ability to produce enough insulin. Stem cells stops converting into the hormone insulin.																												
Prevention	Maintaining a normal weight, eating a healthy and varied diet, and exercising regularly																												
Treatment	Lifestyle changes : eating well -helps to manages levels of blood glucose. Diet should include high-fibre, low-fat foods (fruits, vegetables, grains) and avoid refined sugars and carbohydrates. Low glycaemic index foods are also recommended as they help to maintain more stable blood sugar levels. Exercising -lowers blood pressure and helps insulin to work effectively, reducing the risk of heart disease. Physical activity also helps to lower blood sugar levels, as glucose is being used by cells to enact aerobic respiration. Monitoring : testing blood glucose levels regularly allows patients to effectively manages their treatment, ensuring that these levels stay within a healthy range. Treatment : the first step in diabetes treatment is usually e medication called metformin which decreases the amount of glucose produced in the liver, skeletal muscle, and adipose tissue. Overall. This helps to decreases blood sugar levels. as type 2 diabetes progresses, and the pancreas continues to become less effective at producing insulin, patients may also be required to take insulin injections . Insulin signals to cells to take up glucose, and therefore helps to lower blood glucose levels in diabetics. For obese patients suffering from diabetes, bariatric surgery may also help in reducing its effects. This involves reducing the size of the stomach, either by removing a portion of the stomach, or restricting it using a gastric band. This may potentially reset metabolism, essentially curing diabetes.																												
Future direction	An increased understanding of the factors contributing to development of type 2 diabetes would help scientists and public health workers to design better prevention measures and create effective treatment and management strategies. Future research may be conducted into understanding genetic risks and analysing genomes to identify genes which may place an individual at risk of developing the disease. This would allow the tailoring of pre-emptive, personalised medicine, which would be particularly effective as prevention is the best form of treatment. Type 2 diabetes is commonly treated by taking insulin shots . Synthetic production of insulin is an important area of research, particularly in terms of how to make this process faster, more efficient, and less expensive, in order to increase worldwide access. This could involve genetic engineering, cloning, or large-scale synthetic protein production. An example of current research in this area is the 2017 Sydney University IGEM team, who aimed to provide an open-source solution to the issue of insulin access globally. Another treatment option for diabetes is whole-organ pancreas transplantation , a surgical procedure in which an entirely new, functional pancreas is placed into a patient in order to cure the disease. This option may be developed by research into creation of artificial pancreases to substitute function. This may be facilitated by the development of 3D printing techniques, or growth of pancreatic cells in vitro. Instead of inserting entirely new organs into patients, islet cell transplants may be an option for diabetics. As islet cells are dysfunctional in diabetic patients, unable to properly produce insulin, transplanting these specific cells into patients may represent as a less complex and less invasive option. Research would need to be done into how to grow healthy cells outside the body, and how to insert these into the pancreas in order to restore function																												
Description	a specific kind of skin cancer. It begins in skin cells called melanocytes. Melanocytes produce melanin, the substance that gives your skin colour.																												
Causes	The exact cause is unknown but environmental and genetic factors play a role. Risk factors include Exposure to UV (ultraviolet) radiation. Gene mutation. Fair skin. Family history increases the risk. Weakened immune system																												
Symptoms	The symptoms include: The earliest symptom is typically a change in an existing mole or a new mole arising. The development of a new pigmented or unusual-looking growth on your skin The ABCDE's of melanoma helps identify the disease condition Asymmetry : The mole may not be uniform; one half looks different from the other half. Border irregularity: The moles may be uneven or notched. Colour : The moles may be of different colours and irregular patterns. Diameter : The mole may be bigger in size, at least a 6 mm size. Evolving : The mole may be changing colours, size and texture and might also bleed																												
Host response	DNA repairs will try to repair the damaged DNA. Antibodies are produced to fight the melanoma cells, but they cannot cope for long time.																												
Prevention																													
Treatment	Surgery involves removing the tumour and the skin around the tumour to ensure that none of cancerous cells are left behind, to reduce the risk of a recurrence of the melanoma. Advanced melanomas are treated by radiation, chemotherapy, targeted therapies, and immunotherapy. The type of treatment used will depend on numerous factors, including how far the melanoma has progressed, and the person's age and health. Radiation is when cells are exposed to X-rays, normal cells can repair their DNA more easily than cancer cells can. X-rays damage the DNA of the cancer cells and kill them. Care must be taken in directing the radiation to the specific cells that are intended for destruction. This radiation can be administered externally or by placing a radioactive source in the body near the cancer cells. Chemotherapy drugs slow the growth of cancer cells, but they have not been particularly effective in treating melanoma, and so are not used to treat this form of cancer as much as in other cancers. Targeted therapies involve the use of drugs that affect the molecules that control that control the growth of the tumour cells, which effectively stops the growth and spread of the tumour to other organs. Melanoma cells have mutations that cause the uncontrolled division of cells, leading to the growth of tumours. Targeted therapy drugs interrupt the pathways that cause this uncontrolled division of melanoma cells. Immunotherapy treatment effectively cause the body's own immune system to fight the melanoma. Two approaches are in the early stages of development and use. One of the approaches is to use drugs called checkpoint inhibitors that cause the immune system to recognise and destroy melanoma cells. Cancer cells cause the immune system to ignore them. Checkpoint inhibitors reverse this effect and have proved successful in a number of patients. The other approach is to use vaccinations as a method of treatment rather than prevention. An antigen is produced using the melanoma cells, and when injected allows the immune system to identify and destroy the melanoma cells more easily.																												
Future directions	Future research is required to further develop targeted therapies and immunotherapy treatments for melanoma. The use of targeted therapies is based on the different types of mutations that cause the uncontrolled cell division that occurs in melanoma. Drugs have been developed to interrupt these specific pathways. Scientists are aware that numerous other mutations are yet to be identified, and so research is continuing in order to identify these. Once these mutations are identified, further drugs will then be developed to interrupt many more of these pathways. Currently there are only two immunotherapy drugs available for use and these are not successful on all patients with melanoma. Further research is required to develop a greater variety of immunotherapy drugs, so that the majority of patients can benefit from this form of treatment of melanoma. The use of vaccines to treat melanoma is																												

Incidence	Almost 1 million Australian adults (5.3% of those aged 18 and over) had type 2 diabetes in 2017–18, according to self-reported data from the Australian Bureau of Statistics (ABS) 2017–18 National Health Survey. Proportions were slightly higher for men than women (6.1% and 4.6%, respectively). Age-specific rates for men were higher than women from age 45 onwards. relatively similar across Major cities (4.8%), Inner regional (4.2%) and Outer regional and remote areas (6.0%). around twice as high among those living in the lowest socioeconomic areas (7.0%) compared with the highest socioeconomic areas (3.3%).
Prevalence	The number of people living with type 2 diabetes in Australia increased almost 3-fold between 2000 and 2020, from 399,400 to 1,164,900. After adjusting for age, the prevalence of type 2 diabetes increased from 2.1% in 2000 to 4.0% in 2014. The prevalence rate has decreased slightly since 2014 to 3.8% in 2020.
Mortality rate	Diabetes death rates have remained relatively stable over the last 2 to 3 decades, both where diabetes is the underlying cause of death and where it is the underlying or associated cause of death. On average: diabetes was the underlying cause of death in around 3,300 deaths per year between 1985 and 2018, equating to death rates of 17–22 deaths per 100,000 population for males and 12–15 per 100,000 population for females. Diabetes was the underlying or associated cause of death in around 13,300 deaths per year between 1997 and 2018, equating to death rates of 67–78 per 100,000 population for males and 41–49 per 100,000 population for females.



Mortality rates

	in its early stages. More research is required to further refine this process and make it consistently effective. Further research may also involve investigating the relationship between melanoma and other cancers.
Incidence	In 2018, the age-standardised incidence rate was 54 cases per 100,000 persons (66 for males and 44 for females). In 2022, it is estimated that the age-standardised incidence rate will be 57 cases per 100,000 persons (69 for males and 47 for females). The incidence rate for melanoma of the skin is expected to increase with age, highest for those aged 85–89 years.
Prevalence	At the end of 2017, there were 14,584 people living who had been diagnosed with melanoma of the skin that year, 62,881 people living who had been diagnosed with melanoma of the skin in the previous 5 years (from 2013 to 2017) and 207,943 people living who had been diagnosed with melanoma of the skin in the previous 36 years (from 1982 to 2017).
Mortality rate	In 2020, the age-standardised mortality rate was 4.3 deaths per 100,000 persons (6.4 for males and 2.6 for females). In 2022, it is estimated that the age-standardised mortality rate will be 3.7 deaths per 100,000 persons (5.2 for males and 2.4 for females). The mortality rate for melanoma of the skin is expected to increase with age.

Incidence

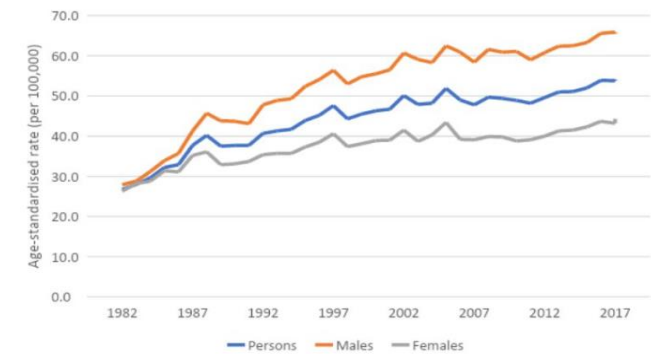


Figure 2. Age-standardised incidence rates for melanoma of the skin, 1982 to 2018, by sex

Mortality

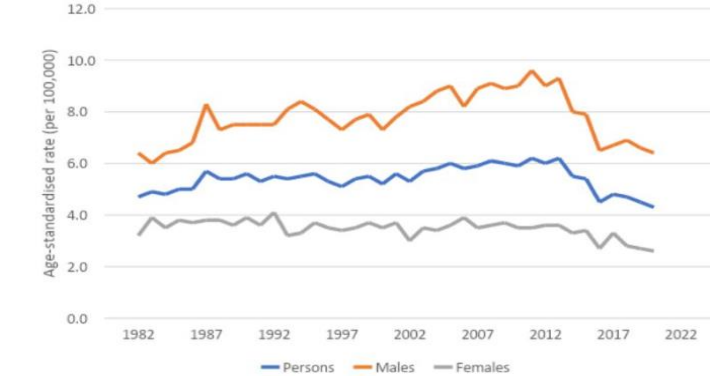
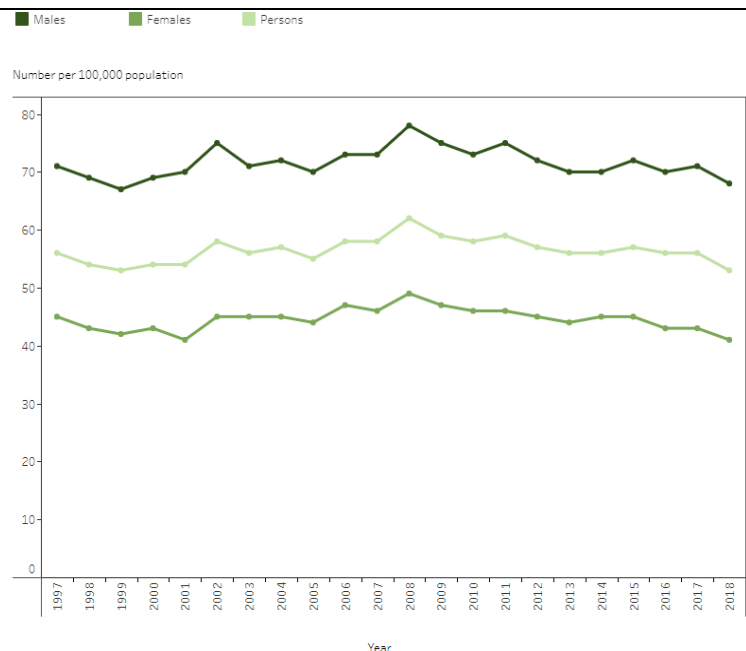


Figure 4. Age-standardised mortality rates for melanoma of the skin, 1983 to 2020, by sex



8.3.4 evaluate the method used in an example of an epidemiological study-epidemiology is the study of patterns, distribution of diseases in a population and how to prevent these diseases. **Mortality** is the number of deaths in a population. **Morbidity** is the number of illnesses in the population. **Prevalence** is the number of new cases and pre-existing cases within a specific time. **Incidence** is the number of new cases in a specific time. **Analytical studies** are planned investigations that test a specific hypothesis (e.g.: comparing the lifestyles of people with a particular disease to determine if they have any risk factors in common). **Descriptive studies** show patterns in the way that a disease is distributed in population (e.g.: where a disease occurs in the world or which groups in a city suffer from certain disease). **Intervention studies** act to measure the effectiveness and safety of certain interventions (e.g.: the effectiveness of a new drug treatment in curing a disease). E.g. analytical study by Richard Doll and Austin Bradford Hill-**smoking and carcinoma** of the lung. **Initial hypothesis**: that the significant increase in deaths attributed to cancer of lung in England and Wales between 1922-1947 (from 612 to 9287 per annum, a roughly fifteenfold increase) was as a result of either atmospheric pollution from car exhaust fumes, surface dust of tarred roads, gas works, industrial plants and coal fires, or the smoking of tobacco, both of which had become more prevalent in the 50 years prior to the study. **Methodology**: 20 London hospitals in the north-west area were asked to participate in the study. Whenever a patient was admitted to said hospitals presenting with carcinoma of the lung, a researcher would visit the hospital and interview the patient. Four designed researchers conducted all interviews of all patients over the period of the study using a set questionnaire. For each lung-carcinoma patient interviewed, an individual of the same sex and in the same 5-year age group affected by a cancer other than carcinoma was interviewed, using the same questionnaire, in the same hospital at or about the same time. A total of 2475 patients participated in the study. The study was conducted for a period of one year. Smoking habits were assessed by asking whether patients: whether they had smoked at any period of their lives. The ages at which they started and stopped. The amount which they were in the habit of smoking before the onset of illness. The main changes in their smoking history and maximum they had ever been in the habit of smoking. The varying proportions smoked in pipes and cigarettes. Whether or not they inhaled. A smoker was defined as a person who had smoked as much as one cigarette a day for as long as one year. Comparisons were made between smokers and non-smokers, as well as groups with varied amounts of smoking (expressed as cigarettes per day). **Evaluation**: the study had a large sample size (2475 patients), with a clearly identified control group with controlled variables (same age and sex) between the populations. Studies were collected across a large number of hospitals. There was consistent use of the same interviewers over the study, using the exact same questionnaire each time. The questionnaire used attempted to thoroughly investigate the history of patients and collected data not only on smoking habits but on other lifestyle factors. The study was conducted for long period of time. The clear and consistent approach to these surveys meant that the study may be reasonably relied upon to have extracted and represented accurate data. However, reliability may have been improved by taking data from hospitals outside of the immediate London area (e.g., from rural hospitals), so that environmental variables could be more sufficiently taken into account. In addition, a study monitoring individual across a large period of their lives, rather than only after they had developed a disease, may have provided a more statistically sound conclusion. **The benefits of engaging in an epidemiology study**-epidemiology is an interdisciplinary field which combines the expertise of epidemiologists, laboratory technicians, statisticians, doctors, and public health professionals. This means that studies are often comprehensive, and wide-reaching in their implications and applications. The use of a thorough, rigorous methodology to trace the origins of disease ensures that findings remain objective and may at times uncover unexpected findings. E.g., when Doll and Hill first began studying causes of lung cancer, they believed that it was mostly due to either car fumes or new material tarmac. However, they discovered that tobacco smoking was the only significant common factor between patients with carcinoma. Epidemiology is concerned with disease surveillance, investigating outbreaks, and conducting observational studies which help to identify risk factors in disease. By identifying the causes of disease, epidemiology provides public health professionals information so that they can strategize and plan programs and campaigns to help prevent disease. E.g., the Australia Cancer Council is currently undertaking the ABC study (Australian Breakthrough Cancer Study), with an outlook towards designing better preventative measures. By determining the levels of disease

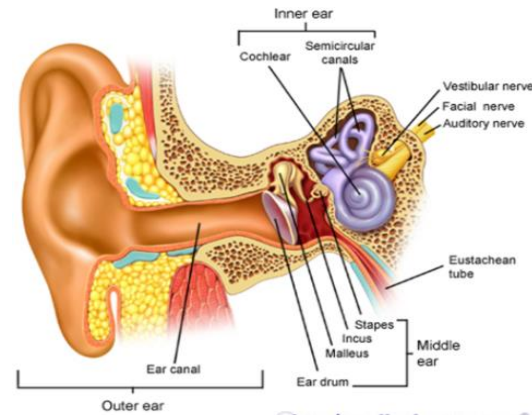
8.4.1 use secondary sources to evaluate the effectiveness of current disease prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to: educational programs and campaigns-e.g., **Slip! Slop! Slap! Seek! Slide!** For skin cancer. **Methods**: television advertisements and community service announcements, including a catchy jingle. Aimed to raise awareness of skin cancer and the risks posed by ultraviolet light exposure from the sun. a cartoon seagull, Sid, appeared in the campaign, encouraging people to "slip on a shirt, slop on sunscreen, slap on hat, seek shade, and slide on sunnies!". The original campaign, launched in the 1980s, was later modified by the SunSmart campaign in 2007, promoting 'Slip! Slop! Slap! Seek! Slide!' in which people were encouraged to also seek shade and slide on sunglasses. Since the success of the program in the 1980s, SunSmart have incorporated further strategies to their campaign, including: implementation of legislative changes. Developing of educational resources for schools and workplaces. Sponsorship of sporting events. **Effectiveness**: one of the most successful health campaigns in Australia's history. Believed to have played a key role in the shift of Australian attitudes towards sun safety. Incidence of two most skin cancers (squamous cell carcinoma and basal-cell carcinoma) has decreased since introduction of the campaign. Incidence of most lethal skin cancer (melanoma) has, however, increased. Program has been shown to be particularly effective amongst younger people. For people aged 20-24, the rate of melanoma has fallen from 25 per 100000 to 14 per 100000 between 1996 and 2010. Estimated to have prevented more than 43000 skin cancers in Victoria (where the campaign was started) between 1988 and 2011. The SunSmart programs are considered to be extremely cost effective, saving the Victoria government \$ 2.20 for every dollar spent in the Victorian public health program. Another e.g., is **beyondblue-secondary schools' program** for mental illnesses (including depression and anxiety). **Methods**: development of a comprehensive curriculum for high-school students to equip them with skills to deal with adverse events (development of "life skills") to help prevent mental health problems. Program encourages development of a sense of self-worth, sense of control, sense of belonging, sense of purpose, sense of future, and sense of humour. Aimed to reduce levels of depression in young people, as well as increased awareness and understanding of depression in adolescents. The program is based on cognitive behavioural therapies. Designed to be delivered over 3 years, with 10 weekly sessions 30-45 minutes in length per year. Consistent delivery, in which principles are built upon, allowing students to develop these skills progressively and concretely. Program delivery based on principles of best-practice. Uses interactive teaching tools such as small group exercises, discussions, role plays, deep-learning tasks, and quizzes. **Effectiveness**: draws on current research conducted nationally and internationally (cognitive behaviour therapies and resilience-based framework). Pre-emptive approach to tackling mental health issues that aims strengthen protective factors, thus reducing the risk of developing mental illness. Targeted towards early high-school students (years 8-10) to address issues before the onset of stressful periods later in life. Displays sound logic in its approach as a result of collaboration between multiple interstate initiatives (Secondary Schools Research Initiative by University of Queensland, Centre for Adolescent Health (Victoria) and South Australian Department of Education and Children's Services). Reported high levels of school engagement with the program between 2003 and

impact at national, regional, and global levels, organisations such as the WHO are able to effectively develop strategies and tools to address global health and inequality. This displays how epidemiology makes an important link between research and public health policy, and the overall benefits of a community-based approach to disease treatment and health. Additionally, identification of the causes of disease allows scientists to understand better where to direct research efforts.

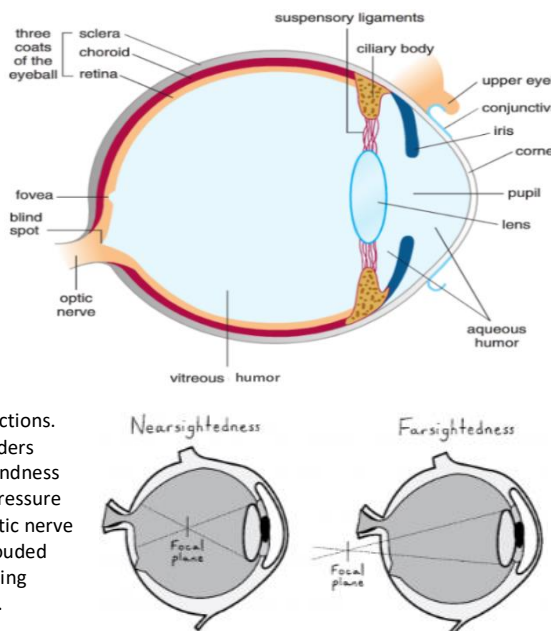
2007. Difficult to assess the lasting impacts of the program due to difficulties in diagnosis of mental illnesses, and the influence of multiple factors contributing to its incidence.

8.4.1 use secondary sources to evaluate the effectiveness of current disease prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to: genetic engineering-many non-infectious diseases are the result of mutations to our genetic material. If we are able to trace the causes of disease to specific changes in our DNA sequences, this opens up the possibility of curing diseases using genetic engineering. Genetic engineering techniques allow us to make changes to the genetic code, thus allowing us to edit sequences causing non-infectious disease. A number of genetic engineering techniques may allow for the prevention of non-infectious disease, including **genetic therapy**-the correction of genetic disorders by introducing a normal, functioning gene into cells, this is achieved by inserting corrected gene sequences into a cell where a defect has occurred. Techniques used to inject new genetic material include the insertion of viral vectors, use of a gene gun, and inorganic nanoparticles. This can be used to treat severe combined **immune deficiency**. **Haemophilia**. **Parkinson's disease**. **CRISPR**-a gene editing system by which point mutations may be accurately introduced into genomes. CRISPR may be used to improve gene therapies by making point mutations (up to 20 bp) to dysfunctional cells. Such changes will alter the genomes of all edited cells, as well as any cells which grow from them, resulting in a lasting somatic cell edit. **Embryo screening or editing**-increased access and decreased costs of whole genome sequencing opens up the possibility of screening embryo for genetic disorders. This may be particularly beneficial for couples who carry known genes for disease. Screening allows selection of embryos only without genetic defects for implantation. Editing technologies, especially at the early embryo stage, open up the possibility that whole organism changes may be made to the offspring.

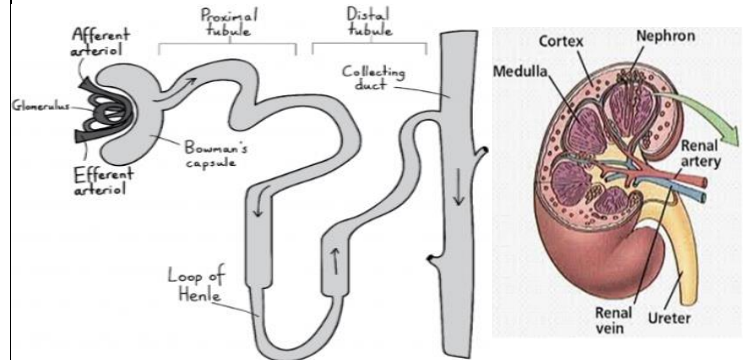
8.5.1 explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example: hearing loss-the ear is composed of three sections: the outer ear (pinna and ear canal). The middle ear (ossicles and ear drum). The inner ear (the cochlea, and auditory nerve and the brain). When sound enters the ear in the form of waves, the **ear drum** vibrates. The vibration movement causes the bones of the middle ear, the **ossicles**, to move in a chain-like fashion. This movement 'knocks' on a membrane window of the **cochlea**. The cochlea is filled with fluid, which moves in response to this knock. Hair cells lining the cochlea are bent in response to the fluid vibrations, which creates an electrical impulse to be sent along the **auditory nerve** and to the brain. **Conductive hearing loss**: occurs when there is damage to the outer or middle ear, resulting in ineffective sound transfer. In these cases, the cochlear may still be functional, but does not receive sufficient signal to create an auditory impulse. Conductive deafness may result from ear infections, otosclerosis (abnormal bone growth in the middle ear), or perforation of the eardrum. **Sensorineural hearing loss**: occurs when there is damage to the inner ear. Sound may arrive at the cochlear, but it is not properly passed on to the auditory nerve, or the auditory nerve itself may be damaged. Damage may be congenital (hearing loss present at birth), as a result of genetic factors or disease, or acquired due to a wide range of factors such as age, noise exposure, physical trauma, or diseases such as meningitis. Hearing loss may be a combination of both conductive and sensorineural hearing loss.



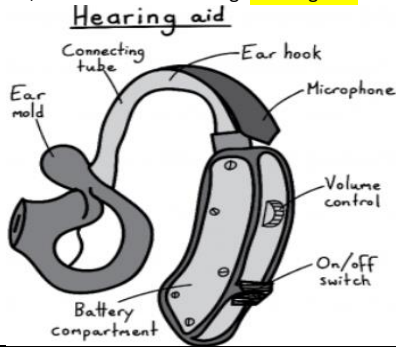
8.5.1 explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example: visual disorders-light enters the eye first through the **cornea**, a clear, curved 'window' on the front of the eye. The cornea refracts light through the pupil, an opening at the front of the eye, the width of which is controlled by the movement of the iris. **Pupils** are able to dilate and contract, the shape of the iris changing how much light enters the eye. Light then passes through the **lens**, a curved, flexible structure which focuses light by shortening or lengthening its width. The globe of the eyeball is filled with **vitreous gel**, a dense transparent substance that gives the eye its spherical shape whilst still allowing light to be transmitted to the back of the eye. Light rays come to a **focal point**, where the image is focused, at the **retina** in the back of the eye. The retina acts like a kind of camera and is composed of two types of cells: rods and cones. These cells are excited by different wavelength of light. As light hits the retina, light energy is converted into electrical impulses by the rods and cones, which send the messages onto the optical nerve, and through to the brain for processing. As a complex organ, there are a number of disorders which may contribute to malfunction of the eye. **Refractive errors**: occur when the eye does not focus light properly, due to incorrect shaping of the cornea. **Myopia**: or near sightedness, occurs when the cornea is too curved or the eyeball is too long, resulting in light refracted at an angle which places the focal plane in front of the retina. In this case, objects far away are blurry. **Hyperopia**: or far-sightedness, occurs when the cornea is too flat or the eyeball is too short, resulting in light being focused beyond the retina. In this case, objects close-up is blurry. **Astigmatism**: is a disorder where vision is blurred at all distances, as a result of a misshapen cornea where curvature is not uniform in all directions. Other prevalent disorders include: **Glaucoma**-blindness due to a build-up of pressure in the eye, causing optic nerve damage. **Cataracts**-clouded areas in the lens, causing blurry or tinted vision.



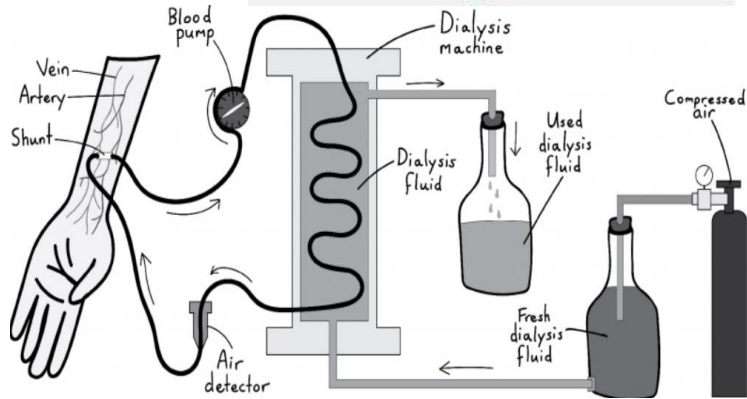
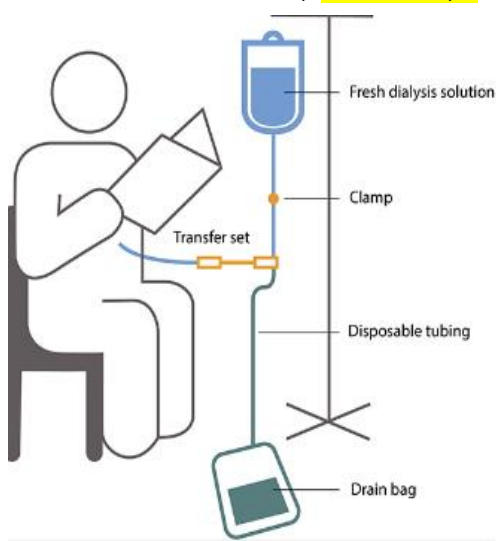
8.5.1 explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example: loss of kidney function-kidneys are the body's cleaners, processing blood to filter out waste products, and balance salt and water levels. Waste products are collected as urine, which moves from the kidneys to the bladder through the ureters, and then out of the body through urethra. Blood enters the kidney from the **renal artery**, and leaves via the **renal vein**. The main areas of the kidney are the cortex (outermost layer of the kidney), medulla (petal-like inner structures), and pelvis (drainage area at centre of kidney, connected to the ureter). Filtering is performed by microscopic structures called **nephrons**, situated across the **cortex** and the **medulla**. Every kidney has about a million nephrons, each performing the job of **filtration**. Capillaries are wound around the nephron structures, providing an interface across which exchange of nutrients may occur. When blood first enters the kidneys, the capillaries are squeezed into a very tightly wound structure called a **glomerulus**. This structure is so compact that all fluid in the blood (everything except RBCs) is squeezed out, into **Bowman's capsule**. In the **proximal convoluted tubule**, essential molecules such as water, glucose, salts and nutrients are transferred back into the capillary. The rest of the nephron structure performs a **balancing function**, regulating the return of salts back into the blood in response to bodily requirements. Hormones may also act on the nephron increasing permeability to certain substances to promote their reabsorption into the blood. All wastes, excess substances and fluids are then drained into the **collecting duct**, which leads to the **renal pelvis** to eventually be drained out of the kidney and the body. Loss of kidney function may result in build-up of wastes, electrolytes, and dangerous levels of fluid in the blood. However, symptoms of kidney disease often have a very late onset. It is possible to lose up to 90% of kidney function before any noticeable symptoms occur. Kidney failure may be as a result of a number of factors, including: diabetes. High blood pressure. Inflammation of important filtration structures. Obstruction of the kidney (i.e., kidney stones or tumours). Infections.



8.5.2 investigate technologies that are used to assist with the effects of a disorder, including but not limited to: hearing loss: cochlear implants, bone conduction implants, hearing aids-**cochlear implants** are electronic devices which replace the function of damaged inner ears (the cochlea). It enables sound, received through an external transmitter, to be transferred to the auditory nerve, allowing individuals who have damaged the hair cells of the inner ear to hear. A sound processor captures sound and turns it into digital code. This digitally coded sound is transmitted to the implant, which converts it to electrical impulses. These impulses are then sent along the electrode, which is placed in the inner ear. The electrode then stimulates the hearing nerve, which sends impulses to the brain to be interpreted as sound. **Bone conduction implants**- this technology is used as an alternative to hearing aids, where individuals have conductive or mixed hearing loss. Sound waves are detected by a processor placed behind the ear. Sounds are converted digitally to vibrations, which are transmitted by the implanted section of the device, through the bone, and the inner ear. Sound vibrations then cause movement in the fluid of the inner ear, stimulating movement of hair cells, and therefore generating electrical impulses. These are then sent along the hearing nerve, as with normal hearing. **Hearing aids**- these devices are worn either in or behind the ear and are used to amplify sound for patients with hearing loss. By magnifying the sound vibrations which enter the ear, hearing sensitivity can be increased. There are two types of hearing aids available, analogue (which convert sound waves into electrical signals, amplify them, then feed them back to the ear) or digital aids are more flexible to individual patient needs and environments.

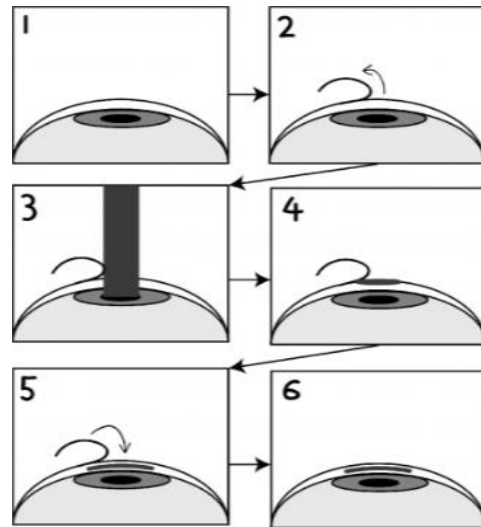


8.5.2 investigate technologies that are used to assist with the effects of a disorder, including but not limited to: loss of kidney function: dialysis-refers to the removal of metabolic wastes, salt, and extra water from the blood by allowing dissolved solutes to pass across a semipermeable membrane. Dialysis is used when kidneys have lost their ability to filter waste products from the blood. It is essential that these products are removed so that the body can continue to function effectively. **Haemodialysis**: waste filtered from the blood outside of the body in a dialyser machine. An A-V fistula is made in the patient's arm, from which blood is drawn, and then passed through a series of semi-permeable membranes. This allows for toxins to be drawn out of the blood into dialyser fluid, and clean blood returned to the body. **Peritoneal dialysis**: cleansing fluid is flown into the peritoneal cavity in the abdomen by a catheter (tube). Wastes are filtered into the cavity from the blood. The fluid with the waste products is then drawn from the abdomen and discarded. Peritoneal dialysis can be performed by yourself at home or whilst travelling, allowing for more flexibility in lifestyle and independence.



8.5.2 investigate technologies that are used to assist with the effects of a disorder, including but not limited to: visual disorders: spectacles, laser surgery-**spectacles**-glasses correct vision by changing the angle at which light hits the cornea, adjusting for misshapen corneas which cause the focal point to deviate from the norm. this allows the eye to focus light on the right spot in the eye, on the retina. Glasses are available with a prescription, which means that they are made for each individual person depending on their needs. For short-sightedness, where the focal point falls in front of the retina, concave lenses allow light to focus on the retina. For far-sightedness, where the focal point falls behind the retina, convex lenses allow light to focus properly on the retina.

Laser surgery-this is used to re-shape the cornea in order to correct vision problems. The surgeon will create a thin flap in the top of the cornea, which is folded back so that the underlying stroma can be accessed. An excimer laser uses short-wave ultraviolet light to remove tiny amounts of tissue. This reshapes the cornea so that it focuses light more accurately (near-sightedness = flatten cornea, far-sightedness = increase curve of cornea).



8.5.3 evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder-**kidney failure**

Technology	Haemodialysis
Description	Haemodialysis is a type of dialysis in which waste is filtered outside of the body using a dialyser machine (or 'artificial kidney'). An A-V fistula is made across the patient's vein, from which blood is drawn. Blood is circulated through a dialyser, through a series of semipermeable membranes, allowing for toxin to be filtered from the blood by passive transport. Dialysers have: a compartment for blood. A compartment for dialysate fluid (which is especially balanced in order to allow excess wastes to pass into it by passive transport along the concentration gradient). A semipermeable membrane separating the two liquids.
Effectiveness	Positive effects : allow for removal of metabolic wastes (urea) when kidneys no longer function (a build-up of urea, known as uraemia, may lead to seizure or a coma, and ultimately death). Provides a long-term solution (people are able to be treated using dialysis for many years) and gives patients time when waiting for kidney transplants. Most hospitals are able to offer dialysis, so access in metropolitan areas is quite good. Negative effects : works only by passive, not active transport, so not all wastes can be filtered, as they will only move into dialysis until equilibrium is reached across the membrane. Requires repeated, large blocks of time (3-4-hour sessions in hospital a few times a week), particularly in cases of advanced kidney failure, so can have a large impact upon patient quality of life, both in terms of time consumption and the requirement to travel to hospital regularly. Limits patient mobility and may be expensive. Dialysis requires that diet and fluid consumption are regulated, and often restricted. Dialysis does not cure kidney disease; it merely mitigates its effects, so if patients have severe kidney failure, they may need dialysis for the rest of their lives, or until they have a kidney transplant.
Summary	Despite the downfalls of haemodialysis, it is evident that this technology is life-saving, and essential for many patients with kidney failure. Ultimately, the technology preserves some quality of life, and extends life expectancy in the case of chronic kidney failure. Although the technology is not a cure for kidney failure, it may be used to tandem with kidney transplants, giving patients more time as they wait for donor organs to become available. Individual patients' conditions may determine whether haemodialysis or peritoneal dialysis is used. The widespread availability of both these technologies in Australia means that there are multiple avenues for treatment of patients with kidney failure.